



# Advanced Time Series Analysis and Forecasting

## Chapter 8: Advanced volatility modelling: realised measures, HAR and multivariate GARCH



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## Acronyms used in this material (1/3)

- ▣ AI: Artificial Intelligence
- ▣ AR: AutoRegressive (model); in event studies: Abnormal Return
- ▣ ARCH: AutoRegressive Conditional Heteroskedasticity
- ▣ ARMA: AutoRegressive Moving Average (model)
- ▣ BEKK: Baba–Engle–Kraft–Kroner (multivariate GARCH model)
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BIC: Bayesian Information Criterion
- ▣ BNHLS: Barndorff–Nielsen–Hansen–Lunde–Shephard
- ▣ BNR: Banca Națională a României (the National Bank of Romania)
- ▣ BV: Bipower Variation
- ▣ CAC: Cotation Assistée en Continu (CAC 40 index, Paris) — the CAC 40 index of the Paris bourse
- ▣ CCC: Constant Conditional Correlation (model)
- ▣ CLT: Central Limit Theorem
- ▣ DCC: Dynamic Conditional Correlation
- ▣ DCC-NL: DCC with Nonlinear shrinkage of the target (Engle–Ledoit–Wolf 2019)
- ▣ DM: Diebold–Mariano (test)
- ▣ EGARCH: Exponential GARCH
- ▣ EODHD: EOD Historical Data (market-data provider)
- ▣ ES: Expected Shortfall
- ▣ FDR: False Discovery Rate
- ▣ FRED: Federal Reserve Economic Data
- ▣ FTSE: Financial Times Stock Exchange (FTSE Russell, index provider)
- ▣ FWER: Family-Wise Error Rate
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GARCH-MIDAS: GARCH with a Mixed Data Sampling long-run component (Engle–Ghysels–Sohn 2013)
- ▣ GARCH-X: GARCH with eXogenous variables

## Acronyms used in this material (2/3)

- ▣ GJR: Glosten–Jagannathan–Runkle (asymmetric GARCH)
- ▣ GMM: Generalized Method of Moments
- ▣ GMV: Global Minimum-Variance (portfolio)
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HAR: Heterogeneous AutoRegressive (model)
- ▣ HAR-CJ: HAR with Continuous and Jump components
- ▣ HARQ: HAR with realised Quarticity
- ▣ HEAVY: High-frEQUENCY-bAsed VolatilitY (models)
- ▣ IGARCH: Integrated GARCH
- ▣ INDPRO: Industrial production index (FRED series code)
- ▣ IQ: Integrated Quarticity
- ▣ LLM: Large Language Model
- ▣ MAE: Mean Absolute Error
- ▣ MCS: Model Confidence Set
- ▣ MFM: Modelling Financial Markets
- ▣ MIDAS: Mixed DAta Sampling (regression)
- ▣ ML: Machine Learning
- ▣ MSE: Mean Squared Error
- ▣ MSTR: MicroStrategy (Strategy Inc.), NASDAQ ticker
- ▣ OLS: Ordinary Least Squares
- ▣ OPG: Outer Product of the Gradient (scores)
- ▣ PPI: Producer Price Index
- ▣ QLIKE: Quasi-LIKElihood loss
- ▣ QML: Quasi-Maximum Likelihood
- ▣ QMLE: Quasi-Maximum Likelihood Estimator
- ▣ QV: Quadratic Variation

## Acronyms used in this material (3/3)

- ▣ RGARCH: Realized GARCH
- ▣ RK: Realised Kernel
- ▣ RMSE: Root Mean Squared Error
- ▣ RQ: Realised Quarticity
- ▣ RV: Realised Variance
- ▣ SV: Stochastic Volatility
- ▣ TQ: Tripower Quarticity
- ▣ TSA: Time Series Analysis
- ▣ TSLA: Tesla Inc., NASDAQ ticker
- ▣ TSRV: Two-Scales Realised Variance
- ▣ US: United States
- ▣ UTC: Coordinated Universal Time (Temps Universel Coordonné)
- ▣ VEC: VEC multivariate GARCH (half-vectorised covariance)
- ▣ VIX: CBOE Volatility Index
- ▣ VR: Variance Ratio
- ▣ WPSFD49207: PPI by commodity: final demand, finished goods (FRED series code)

## Today's question and route

- ▣ **Question:** how precisely can we measure, model and forecast a variance that is never observed, and which of our tools still work when the data are noisy, jumpy or high-dimensional?
  - ▶ two routes to volatility: a parametric filter of daily returns (GARCH) and a nonparametric measurement from intraday prices (realised measures); the frontier combines them
- ▣ **Route** of the chapter
  - ▶ quasi-maximum likelihood for GARCH and robust inference; long-run components: component GARCH and GARCH-MIDAS with macroeconomic drivers
  - ▶ realised measures: asymptotics of realised variance, microstructure noise, realised kernels, bipower variation and jump tests
  - ▶ forecasting with realised measures: HAR, HARQ, Realized GARCH, HEAVY; robust loss functions
  - ▶ multivariate: BEKK, DCC and cDCC, the curse of dimensionality and DCC-NL, realised covariance
- ▣ We build on [TSA, Chapter 5](#) (GARCH, GJR, EGARCH, news impact, QLIKE) and [TSA, Chapter 14](#) (DCC basics); MFM, Chapters 5–9 apply these tools to market risk; Seminar 8 comes before this lecture

## Learning outcomes

- ▣ State the conditions under which Gaussian QML for GARCH is consistent and asymptotically normal, and compute Bollerslev–Wooldridge standard errors
- ▣ Separate short-run from long-run volatility with component GARCH and GARCH-MIDAS, and judge whether a macroeconomic driver matters
- ▣ Derive the limit theory of realised variance, explain the effect of noise and jumps, and choose among RV, realised kernels and bipower variation
- ▣ Forecast volatility with HAR, HARQ and Realized GARCH, and compare forecasts with losses that are robust to a noisy proxy
- ▣ Estimate DCC and cDCC models, explain why large covariance matrices need shrinkage, and measure correlation from intraday data

## Reading, data and tools

- Backbone: [Francq and Zakoïan \(2019\)](#); [Andersen, Bollerslev, Diebold and Labys \(2003\)](#); [Barndorff-Nielsen and Shephard \(2002\)](#); [Barndorff-Nielsen, Hansen, Lunde and Shephard \(2008\)](#); [Corsi \(2009\)](#); [Hansen, Huang and Shek \(2012\)](#); [Engle \(2002\)](#); [Patton \(2011\)](#)
  - ▶ case studies: [Engle, Ghysels and Sohn \(2013\)](#) (GARCH-MIDAS), [Bollerslev, Patton and Quaedvlieg \(2016\)](#) (HARQ), [Hansen, Huang and Shek \(2012\)](#) (Realized GARCH), [Engle, Ledoit and Wolf \(2019\)](#) (DCC-NL)
- Python Quantlets of this chapter: [Quantlets/Ch\\_08](#)
  - ▶ QML with sandwich standard errors, component GARCH, GARCH-MIDAS, realised kernels, jump tests, HAR/HARQ, Realized GARCH, DCC/cDCC and nonlinear shrinkage written out in `numpy`
- Lecture notebook: [open in Google Colab](#)

## Data used in this chapter

| Series                                                                                                            | Source                                                                               | Use                |
|-------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|--------------------|
| Daily realised measures of six equity indices (5-minute RV, BV, realised kernel, open-to-close return), 2000–2022 | <a href="#">Heber, Lunde, Shephard and Sheppard (2009)</a> , v0.3                    | HAR, RGARCH        |
| Bitcoin and Ether: one-minute prices 2018–2026, one-second prices August 2026                                     | Binance public data ( <a href="#">data.binance.vision</a> ); measures computed by us | noise, jumps, HARQ |
| S&P 500, DAX, BET, Bitcoin, 14 US stocks and 3 equity ETFs, daily                                                 | EODHD                                                                                | QML, MIDAS, DCC    |
| EUR/RON reference rate                                                                                            | BNR                                                                                  | QML                |
| US industrial production, PPI finished goods (monthly)                                                            | FRED (INDPRO, WPSFD49207)                                                            | GARCH-MIDAS        |

- The Oxford-Man library ended in February 2022 (archived copy, cited as its terms require); Binance one-minute data extend the realised measures to 18 September 2026; a calibrated simulation gives the theory a known truth
- RV: realised variance; BV: bipower variation; PPI: producer price index

## From the trading floor to the tick



New York Stock Exchange, 1908

Photo: Helen D. Van Eaton, *Leslie's Monthly Magazine* (1908);  
public domain; [Wikimedia Commons](#)

- 1982–1986: ARCH and GARCH Engle (1982), Bollerslev (1986): volatility as a filtered conditional variance of daily returns
- 1992–2004: QML theory Bollerslev and Wooldridge (1992), Lee and Hansen (1994), Lumsdaine (1996), Francq and Zakoïan (2004); components Engle and Lee (1999); DCC Engle (2002)
- 1998–2008: realised variance Andersen and Bollerslev (1998), Andersen, Bollerslev, Diebold and Labys (2003), Barndorff-Nielsen and Shephard (2002); bipower variation Barndorff-Nielsen and Shephard (2004); noise and kernels Zhang, Mykland and Aït-Sahalia (2005), Barndorff-Nielsen, Hansen, Lunde and Shephard (2008)
- 2009–2019: HAR Corsi (2009), Realized GARCH Hansen, Huang and Shek (2012), GARCH-MIDAS Engle, Ghysels and Sohn (2013), HARQ Bollerslev, Patton and Quaedvlieg (2016), DCC-NL Engle, Ledoit and Wolf (2019)
- Today: high-frequency data are cheap; the questions are statistical (noise, jumps, dimension, evaluation)

## From TSA to this chapter

- Known ([TSA, Chapter 5](#)): GARCH(1,1), GJR, EGARCH, the news impact curve, ML with Gaussian and Student- $t$  errors, QLIKE; ([TSA, Chapter 14](#)): CCC and DCC
- New here
  - ▶ why Gaussian ML still works when returns are not Gaussian, and which standard errors are then valid
  - ▶ volatility with a slow, macro-driven component; volatility measured from intraday prices, with its own sampling error
  - ▶ models that use realised measures; evaluation against a noisy target; covariance matrices with many assets
- Stochastic volatility (a latent log-variance with its own shock): [Chapter 6](#); Markov-switching GARCH: [Chapter 7](#); VaR and ES backtesting: [Chapter 9](#); rough volatility: [Chapter 10](#)

## The Gaussian quasi-likelihood (1/2): the model

- The return shock is a volatility times a standardised shock; the variance follows a GARCH(1,1) recursion

$$\varepsilon_t = \sigma_t(\theta_0) \eta_t, \quad \sigma_t^2(\theta) = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2(\theta)$$

- Notation

- ▶  $\varepsilon_t$ : the demeaned return of day  $t$ ;  $\sigma_t^2(\theta)$ : its variance conditional on the past  $\mathcal{F}_{t-1}$  (the information up to day  $t-1$ )
  - ▶  $\eta_t$ : i.i.d. standardised shock,  $\mathbb{E}\eta_t = 0$ ,  $\mathbb{E}\eta_t^2 = 1$ ; its law (Normal, Student- $t$ , skewed) is unknown
  - ▶  $\theta = (\omega, \alpha, \beta)$ , true value  $\theta_0$ :  $\omega > 0$  sets the level,  $\alpha \geq 0$  the reaction to yesterday's squared shock,  $\beta \geq 0$  the persistence of yesterday's variance
- The recursion starts from an arbitrary  $\sigma_1^2$ ; the effect of the start vanishes geometrically when  $\beta < 1$

## The Gaussian quasi-likelihood (2/2): estimator and score

- QMLE: maximise the Gaussian log-likelihood, used as an estimating equation even if  $\eta_t$  is not Gaussian

$$\hat{\theta} = \arg \max_{\theta} \sum_{t=1}^T \ell_t(\theta), \quad \ell_t(\theta) = -\frac{1}{2} \left( \ln \sigma_t^2(\theta) + \frac{\varepsilon_t^2}{\sigma_t^2(\theta)} \right)$$

►  $\ell_t$ : the Gaussian log-density of day  $t$  without its constant;  $T$ : number of days;  $\hat{\theta}$ : the estimate

- The score, the gradient of  $\ell_t$ , has conditional mean zero at the true value

$$s_t(\theta) = \frac{\partial \ell_t}{\partial \theta} = \frac{1}{2} \left( \frac{\varepsilon_t^2}{\sigma_t^2} - 1 \right) \frac{1}{\sigma_t^2} \frac{\partial \sigma_t^2}{\partial \theta}$$

► at  $\theta_0$ ,  $\varepsilon_t^2 / \sigma_t^2 = \eta_t^2$ , hence  $\mathbb{E}(s_t | \mathcal{F}_{t-1}) = 0$  because  $\mathbb{E}\eta_t^2 = 1$

► only the conditional variance must be right; the shape of the law of  $\eta_t$  is never used

- The score is a martingale difference (conditional mean zero given the past)

► a CLT for martingales gives asymptotic normality without independence of  $\varepsilon_t$

## Consistency and asymptotic normality (1/2): stationarity and consistency

- GARCH(1,1) is strictly stationary if and only if the log of the daily variance multiplier has a negative mean

$$\mathbb{E} \ln(\alpha_0 \eta_t^2 + \beta_0) < 0$$

- ▶  $\alpha_0 \eta_t^2 + \beta_0$ : the factor by which a shock to the variance is carried from one day to the next;  $\alpha_0, \beta_0$ : true values
- ▶ by Jensen's inequality, weaker than  $\alpha_0 + \beta_0 < 1$  (finite variance): IGARCH ( $\alpha_0 + \beta_0 = 1$ ) is strictly stationary, and  $\mathbb{E} \varepsilon_t^2$  may be infinite
- ▶ Lee and Hansen (1994), Lumsdaine (1996): GARCH(1,1), including IGARCH; Berkes, Horváth and Kokoszka (2003), Francq and Zakoïan (2004): GARCH( $p, q$ ) under strict stationarity
- Consistency of  $\hat{\theta}$  Francq and Zakoïan (2004)
  - ▶ conditions: strict stationarity,  $\theta_0$  in a compact parameter set, identifiability ( $\eta_t^2$  is not a constant)
  - ▶ no moment of  $\varepsilon_t$  is needed
- Even explosive GARCH ( $\mathbb{E} \ln(\alpha_0 \eta_t^2 + \beta_0) > 0$ ):  $(\hat{\alpha}, \hat{\beta})$  remain consistent and asymptotically normal,  $\omega$  is not identified Jensen and Rahbek (2004)

## Consistency and asymptotic normality (2/2): the limit law

- With  $\theta_0$  interior to the parameter set and a finite fourth moment of  $\eta_t$ , the estimator is asymptotically normal

$$\sqrt{T}(\hat{\theta} - \theta_0) \rightarrow N(0, (\kappa_\eta - 1)J^{-1}), \quad J = \mathbb{E} \left[ \frac{1}{\sigma_t^4} \frac{\partial \sigma_t^2}{\partial \theta} \frac{\partial \sigma_t^2}{\partial \theta'} \right]$$

- ▶  $\kappa_\eta = \mathbb{E}\eta_t^4$ : kurtosis of the shock (3 for the Normal distribution);  $\kappa_\eta - 1 = \text{Var}(\eta_t^2)$
- ▶  $J$ : a  $3 \times 3$  matrix, the mean outer product of the relative sensitivities  $\sigma_t^{-2} \partial \sigma_t^2 / \partial \theta$ ;  $\rightarrow$ : convergence in distribution as  $T \rightarrow \infty$
- ▶ reading: the sampling variance of  $\hat{\theta}$  grows linearly with the kurtosis of the shocks
- Boundary: if  $\alpha_0 = 0$ ,  $\theta_0$  is not interior and the limit is the projection of a normal vector on a cone
  - ▶ this is why [TSA, Chapter 5](#) tests ARCH effects one-sided

## The sandwich and the Bollerslev–Wooldridge standard errors (1/2)

- General QML result: the asymptotic variance is a “sandwich” of two matrices

$$\sqrt{T}(\hat{\theta} - \theta_0) \rightarrow N(0, A^{-1} B A^{-1}), \quad A = -\mathbb{E} \frac{\partial^2 \ell_t}{\partial \theta \partial \theta'}, \quad B = \mathbb{E} s_t s_t'$$

- ▶  $A$ : the expected negative Hessian (curvature of the log-likelihood);  $B$ : the variance of the score  $s_t$
  - ▶ if the likelihood is the true one,  $A = B$  (information equality) and the variance reduces to  $A^{-1}$ ; under QML,  $A \neq B$
- For GARCH both matrices are proportional to  $J$  (▶ [Appendix](#))

$$A = \frac{1}{2} J, \quad B = \frac{\kappa_\eta - 1}{4} J, \quad A^{-1} B A^{-1} = (\kappa_\eta - 1) J^{-1}$$

- ▶ the sandwich recovers the limit law of the previous slide, for any law of  $\eta_t$  with  $\kappa_\eta < \infty$

## The sandwich and the Bollerslev–Wooldridge standard errors (2/2)

### □ Three ways to compute standard errors (s.e.)

- ▶ Hessian only:  $A^{-1} = 2J^{-1}$ , correct only if  $\kappa_\eta = 3$ ; too small by the factor  $\sqrt{(\kappa_\eta - 1)/2}$  when  $\kappa_\eta > 3$
- ▶ outer product of the scores (OPG):  $B^{-1} = \frac{4}{\kappa_\eta - 1} J^{-1}$ , even smaller when  $\kappa_\eta > 3$
- ▶ sandwich Bollerslev and Wooldridge (1992):  $\hat{A}^{-1} \hat{B} \hat{A}^{-1}$ , valid whatever the law of  $\eta_t$  (with  $\kappa_\eta < \infty$ )

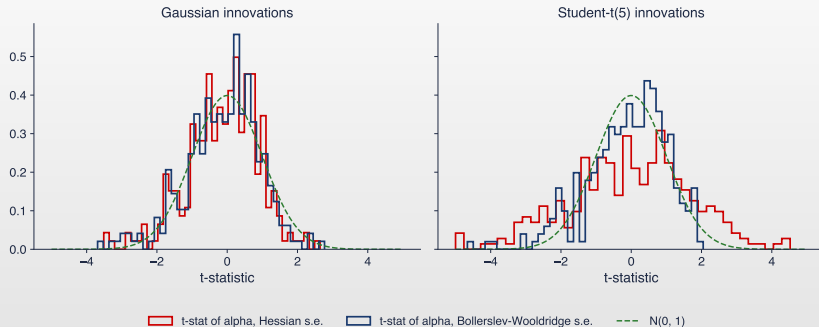
### □ Estimates used by Bollerslev and Wooldridge

- ▶  $\hat{A}$ : the observed Hessian of  $-\frac{1}{T} \sum_t \ell_t$  at  $\hat{\theta}$ ;  $\hat{B} = \frac{1}{T} \sum_t \hat{s}_t \hat{s}_t'$ , with  $\hat{s}_t = s_t(\hat{\theta})$  the estimated score

### □ Robust Wald and score tests use the sandwich

- ▶ the quasi-likelihood-ratio (quasi-LR) statistic is no longer  $\chi^2$ , but a weighted sum of  $\chi_1^2$  variables (chi-square with one degree of freedom)

## Hessian against sandwich: a Monte Carlo

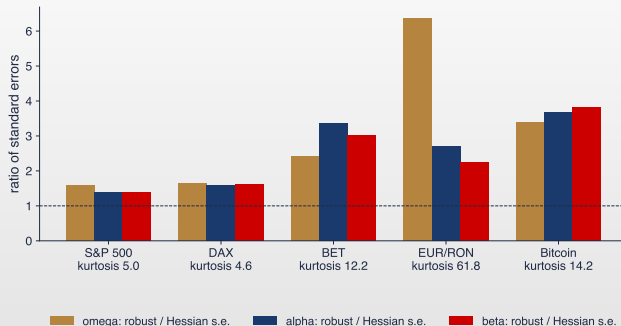


- GARCH(1,1) with  $(\omega, \alpha, \beta) = (0.05, 0.08, 0.90)$ ,  $T = 2000$ , 300 replications; Gaussian QML in each; histogram of  $(\hat{\alpha} - \alpha_0)/\hat{s.e.}$

## Interpreting the Monte Carlo

- ▣ Gaussian  $\eta$ : coverage of nominal 95% intervals for  $\alpha$  is 95.0% (Hessian) and 94.3% (sandwich): both fine
- ▣ Student- $t(5)$ ,  $\kappa_\eta = 9$ : Hessian intervals cover 73.0% for  $\alpha$  and 74.7% for  $\beta$ ; sandwich intervals 91.7% and 92.3%
- ▣ The theory predicts the factor  $\sqrt{(9-1)/2} = 2$  between the two standard errors; the point estimates are the same
- ▣ With fat tails, a Hessian-based  $t$ -test of  $\alpha$  rejects far too often: report sandwich standard errors by default

## Robust and naive standard errors in five markets



- Gaussian QML of GARCH(1,1), daily returns 2010–2026; bars: sandwich s.e. divided by Hessian s.e.; labels: kurtosis  $\hat{\kappa}_\eta$  of the standardised residuals

## Interpreting the five markets

- ▣ S&P 500:  $\hat{\alpha} = 0.154$  with s.e. 0.013 (Hessian) against 0.018 (sandwich);  $\hat{\kappa}_{\eta} = 5.0$ , theoretical factor 1.42
- ▣ BET:  $\hat{\kappa}_{\eta} = 12.2$ , the sandwich s.e. of  $\hat{\alpha}$  is 3.36 times larger; EUR/RON (managed float): 2.71 times,  $\hat{\kappa}_{\eta} = 61.8$
- ▣ The ratios track  $\sqrt{(\hat{\kappa}_{\eta} - 1)/2}$  only roughly: in the data  $\eta_t$  is not i.i.d., and the sandwich does not need it to be
- ▣ Student- $t$  ML raises the log-likelihood by 124.4 (S&P 500) and 233.0 (BET), but is consistent only if the  $t$  shape is right; QML needs only the variance equation

## Recap: QML for GARCH

- ▣ Gaussian QML is consistent if the variance equation is right, whatever the shape of the innovations
- ▣ Strict stationarity, not finite variance, is what the theory needs; IGARCH and even explosive GARCH are covered
- ▣ With fat tails, Hessian standard errors are too small by  $\sqrt{(\kappa_\eta - 1)/2}$ ; use Bollerslev–Wooldridge

## GARCH with exogenous variables

- GARCH-X: yesterday's value of an observed variable enters the variance equation

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2 + \pi' x_{t-1}$$

- ▶  $x_{t-1}$ : vector of exogenous variables known at the end of day  $t - 1$ ;  $\pi$ : their loadings ( $\pi'x$ : the weighted sum)
- ▶ positivity of  $\sigma_t^2$ :  $x_{t-1} \geq 0$  and  $\pi \geq 0$ , or a specification in logs
- Candidates for  $x_t$ : a realised measure (leads to HEAVY and Realized GARCH below), the VIX, a macro or policy-uncertainty index, trading volume
  - ▶ QML theory carries over if  $x_t$  is stationary and ergodic; its own dynamics need not be modelled for one-step forecasts
  - ▶ multi-step forecasts need a model for  $x_t$ : this is exactly what Realized GARCH and HEAVY add
- A slow variable (a monthly macro series) cannot enter a daily recursion at its own frequency: one needs a component structure (next slides)

## The component GARCH of Engle and Lee (1/2): the model

- Engle and Lee (1999): the variance is a slowly moving long-run level  $q_t$  plus a transitory deviation from it

$$\sigma_t^2 = q_t + \alpha(\varepsilon_{t-1}^2 - q_{t-1}) + \beta(\sigma_{t-1}^2 - q_{t-1})$$

$$q_t = \omega + \rho q_{t-1} + \phi(\varepsilon_{t-1}^2 - \sigma_{t-1}^2)$$

- ▶  $q_t$ : long-run (permanent) component;  $\rho$ : its persistence, close to 1;  $\omega$ : its intercept
  - ▶  $\sigma_t^2 - q_t$ : transitory component;  $\alpha, \beta$ : its reaction and persistence, with  $\alpha + \beta < \rho$
  - ▶  $\varepsilon_{t-1}^2 - \sigma_{t-1}^2$ : the variance surprise of yesterday (mean zero);  $\phi$ : how much of it moves the long-run level
- Both components are driven by the same shock: the model is a restricted GARCH(2,2)

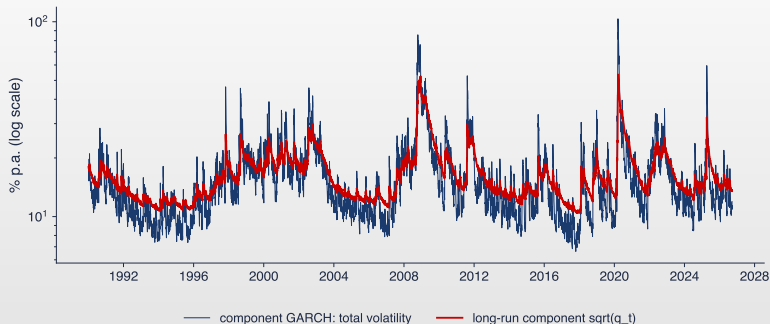
## The component GARCH of Engle and Lee (2/2): two speeds

- Half-life: the number of days after which half of a shock to a component has died out

$$HL_q = \frac{\ln(1/2)}{\ln \rho}, \quad HL_{\sigma-q} = \frac{\ln(1/2)}{\ln(\alpha + \beta)}$$

- ▶  $HL_q$ : half-life of the long-run component;  $HL_{\sigma-q}$ : half-life of the transitory component
- ▶ a persistence  $\rho = 0.99$  gives about 69 days;  $\alpha + \beta = 0.9$  gives about 6.6 days
- ▶ GARCH(1,1) forces a single half-life,  $\ln 0.5 / \ln(\alpha + \beta)$
- Two exponentials approximate a slowly decaying (hyperbolic) autocorrelation of squared returns over a finite range of lags
  - ▶ long memory proper: [Chapter 10](#)

## Component GARCH for the S&P 500



□ Daily returns since 1990,  $T = 9\,240$ ; Gaussian QML; annualised total volatility  $\sqrt{252\sigma_t^2}$  and long-run component  $\sqrt{252q_t}$

## Interpreting the component model

- ▣  $\hat{\rho} = 0.995$ ,  $\hat{\phi} = 0.026$ ; transitory part  $\hat{\alpha} + \hat{\beta} = 0.950$ : half-lives 150 days (long run) and 13.6 days (short run)
- ▣ GARCH(1,1) on the same data: persistence 0.983, one half-life of 41 days, a compromise between the two
- ▣ Quasi-LR statistic against GARCH(1,1): 44.1; under the null  $\phi = 0$  the parameter  $\rho$  is not identified (Davies problem), so the  $\chi^2_2$  reference is only indicative
- ▣ After 2008, 2020 and 2025 the long-run level stays high for months while daily volatility has already fallen: this matters for horizons beyond a few weeks

## GARCH-MIDAS (1/3): a macro-driven long-run component



Eric Ghysels, 2019

Photo: Eghysels (2019); CC BY-SA 4.0;

Wikimedia Commons

- Engle, Ghysels and Sohn (2013): the daily variance is the product of a monthly level  $\tau_t$  and a daily factor  $g_{i,t}$

$$r_{i,t} = \mu + \sqrt{\tau_t g_{i,t}} \eta_{i,t}$$

- ▶  $r_{i,t}$ : return of day  $i$  of month  $t$ ;  $\mu$ : mean return;  $\eta_{i,t}$ : i.i.d. shock with mean 0 and variance 1

- Short run: a GARCH(1,1) with mean 1, re-scaled by  $\tau_t$

$$g_{i,t} = (1 - \alpha - \beta) + \alpha \frac{(r_{i-1,t} - \mu)^2}{\tau_t} + \beta g_{i-1,t}$$

- ▶  $\alpha, \beta$ : reaction and persistence of the daily component, as in GARCH(1,1)

## GARCH-MIDAS (2/3): the long-run component

- Long run: MIDAS (mixed-data sampling) regression of  $\ln \tau_t$  on  $K$  lags of a monthly variable

$$\ln \tau_t = m + \theta \sum_{k=1}^K \varphi_k(w) X_{t-k}, \quad \varphi_k(w) \propto (1 - k/K)^{w-1}$$

- ▶  $X_{t-k}$ : the monthly driver  $k$  months ago;  $m$ : intercept;  $\theta$ : effect of the driver (sign: pro- or countercyclical)
  - ▶  $\varphi_k(w)$ : beta lag weights, normalised to sum to 1 and restricted to decay [Ghysels, Santa-Clara and Valkanov \(2006\)](#);  $w \geq 1$ : the larger  $w$ , the faster the decay;  $w = 1$ : equal weights
- Choices of  $X$ 
    - ▶ realised variance of past months, in levels:  $\tau_t = m + \theta \sum_k \varphi_k(w) \text{RV}_{t-k}$ , with  $\text{RV}_{t-k}$  the sum of squared daily returns of month  $t - k$
    - ▶ macro data: industrial production growth, PPI inflation

## GARCH-MIDAS (3/3): identification, estimation and the variance ratio

- $\tau_t$  is constant within the month and  $\mathbb{E}g_{i,t} = 1$ : the split is identified by the different frequencies, without a filter for a latent variable
  - ▶ Gaussian QML of all parameters  $(\mu, \alpha, \beta, m, \theta, w)$  in one step; sandwich standard errors
- Variance ratio: the share of the variation of log volatility explained by the long-run component

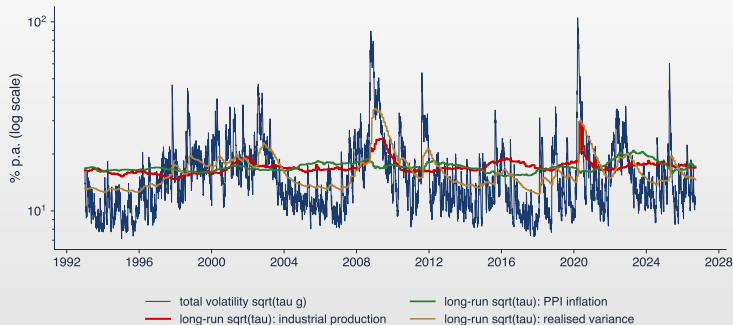
$$\text{VR} = \frac{\text{Var}(\ln \tau_t)}{\text{Var}(\ln(\tau_t g_{i,t}))} \in [0, 1]$$

- ▶ VR close to 0: the long-run component is almost flat; close to 1: it carries most of the variation
  - ▶ a macro variable can be significant ( $\hat{\theta} \neq 0$ ) and still explain little of daily volatility (small VR)
- Inference on  $w$  is nonstandard when  $\theta = 0$  ( $w$  is then not identified) and at the boundary  $w = 1$
- Extensions: two-sided weights, several regressors, a second (daily) component [Conrad and Kleen \(2020\)](#); forecasts beyond one month use only the long-run part

## Case study: Engle, Ghysels and Sohn (2013) on our data

- ▣ **Question** of Engle, Ghysels and Sohn (2013): does the stock market's long-run volatility move with macroeconomic fundamentals (output growth, inflation)?
  - ▶ their sample of US stock returns is much longer than ours; their benchmark long-run component is a fixed-window realised variance
- ▣ **Our replication:** S&P 500 daily returns, common sample from 04.01.1993,  $T = 8482$ ;  $K = 36$  monthly lags; restricted beta weights
  - ▶ three long-run drivers: monthly realised variance (sum of squared daily returns), industrial production growth (FRED INDPRO), PPI inflation (FRED WPSFD49207)
  - ▶ all models against GARCH(1,1) on the same days; QML with sandwich standard errors
- ▣ Extension for a project: Romanian industrial production and the BET (Seminar 8, B5); real-time vintages of the macro data

## GARCH-MIDAS for the S&amp;P 500



□ Annualised  $\sqrt{252\tau_t g_{i,t}}$  of the industrial-production model and the long-run components  $\sqrt{252\tau_t}$  of the three models

## GARCH-MIDAS estimates

| Long-run driver              | $\hat{\alpha}$ | $\hat{\beta}$ | $\hat{\theta}$ (s.e.) | $\hat{w}$ | VR (%) | BIC     |
|------------------------------|----------------|---------------|-----------------------|-----------|--------|---------|
| realised variance (levels)   | 0.119          | 0.841         | 0.0225 (0.0058)       | 5.939     | 25.1   | 22710.0 |
| industrial production growth | 0.110          | 0.870         | -0.551 (0.353)        | 3.692     | 4.5    | 22732.5 |
| PPI inflation                | 0.108          | 0.873         | 0.753 (1.260)         | 1.000     | 2.2    | 22733.3 |
| GARCH(1,1)                   |                |               |                       |           |        | 22721.9 |

- Same days for all models;  $BIC = -2 \ln L + k \ln T$  (lower is better); s.e.: Bollerslev–Wooldridge

## Interpreting GARCH-MIDAS

- ▣ Realised variance as the slow driver:  $\hat{\theta} = 0.0225$  ( $t = 3.86$ ),  $VR = 25.1\%$ , BIC below GARCH(1,1): a long-run component exists
- ▣ Industrial production:  $\hat{\theta} = -0.551$  ( $t = -1.56$ ), the countercyclical sign of [Engle, Ghysels and Sohn \(2013\)](#), but VR only 4.5% and a higher BIC than GARCH(1,1)
- ▣ PPI inflation:  $\hat{w}$  at the boundary 1 (flat weights) and  $\hat{\theta}$  imprecise: no evidence in 1993–2026
- ▣ Replication verdict: the qualitative sign survives, the strength does not; macro effects need long samples with deep recessions, which is why the original study uses a much longer history

## Recap: Long-run components

- ▣ Volatility has at least two speeds; a single GARCH persistence averages them
- ▣ GARCH-MIDAS lets a monthly variable drive the long-run level; identification comes from the mixed frequencies
- ▣ Report VR and BIC next to  $\hat{\theta}$ : significance is not importance

## Prices as Itô semimartingales (1/2): the model

- Within day  $t$ , the log price moves by a drift, a Brownian part scaled by the spot volatility, and jumps

$$dX_s = \mu_s ds + \sigma_s dW_s + dJ_s, \quad s \in [0, 1]$$

- ▶  $X_s$ : log price at intraday time  $s$  (the day is rescaled to  $[0, 1]$ );  $\mu_s$ : drift;  $W_s$ : standard Brownian motion
- ▶  $\sigma_s$ : stochastic spot (instantaneous) volatility;  $J_s$ : a jump process with finitely many jumps per day,  $\Delta J_s$ : the jump at time  $s$

- Quadratic variation of the day: the diffusive part plus the squared jumps

$$QV_t = IV_t + \sum_{s \leq 1} (\Delta J_s)^2, \quad IV_t = \int_0^1 \sigma_s^2 ds$$

- ▶  $IV_t$ : integrated variance, the average of the spot variance over the day
- ▶ without jumps and with  $\sigma$  independent of  $W$ :  $r_t | IV_t \sim N(\int_0^1 \mu_s ds, IV_t)$ , so  $IV$  is the variance that matters for the daily return  $r_t$

## Prices as Itô semimartingales (2/2): realised variance

- Realised variance: the sum of the  $n$  squared intraday returns of the day

$$\text{RV}_t = \sum_{i=1}^n r_{t,i}^2, \quad r_{t,i} = X_{i/n} - X_{(i-1)/n}$$

- ▶  $n$ : number of intraday intervals (78 five-minute returns in a 6.5-hour session);  $r_{t,i}$ : log return over the  $i$ -th interval
- ▶  $\text{RV}_t \rightarrow \text{QV}_t$  in probability as  $n \rightarrow \infty$ : RV measures the total (diffusive plus jump) variation
- Realised variance: the sum of the  $n$  squared intraday returns of the day
- ▶ The drift does not matter in the limit: it is  $O(1/n)$  per interval, while the Brownian part is  $O(n^{-1/2})$
- ▶ volatility becomes observable *ex post*, without a model Andersen, Bollerslev, Diebold and Labys (2003), Barndorff-Nielsen and Shephard (2002)

## The central limit theorem for realised variance (1/2)

- Barndorff-Nielsen and Shephard (2002): without jumps, the error of RV shrinks at rate  $\sqrt{n}$  and is mixed normal

$$\sqrt{n}(\text{RV}_t - \text{IV}_t) \rightarrow MN(0, 2\text{IQ}_t), \quad \text{IQ}_t = \int_0^1 \sigma_s^4 ds$$

- ▶  $\text{IQ}_t$ : integrated quarticity, the daily average of  $\sigma_s^4$ ; it sets the size of the error
  - ▶  $MN$  (mixed normal): normal given the path of  $\sigma$ , with a random variance  $2\text{IQ}_t$
  - ▶ the convergence is *stable* in law: one may divide by a random, consistently estimated  $\sqrt{\text{IQ}_t}$  and keep the limit
- Rate  $\sqrt{n}$ : with  $n = 78$  and volatility constant within the day, the relative error is about  $\sqrt{2/78} \approx 16\%$
- ▶ the measurement error  $\text{RV}_t - \text{IV}_t$  is heteroskedastic (it scales with  $\text{IQ}_t$ ): HARQ, below, is built on this fact

## The central limit theorem for realised variance (2/2): feasible intervals

- ▣ Feasible version: estimate IQ by the realised quarticity RQ, then studentise

$$\widehat{\text{IQ}}_t = \text{RQ}_t = \frac{n}{3} \sum_{i=1}^n r_{t,i}^4, \quad \frac{\text{RV}_t - \text{IV}_t}{\sqrt{\frac{2}{3} \sum_i r_{t,i}^4}} \rightarrow N(0, 1)$$

- ▶ the factor  $\frac{n}{3}$  makes RQ unbiased under constant volatility, because  $\mathbb{E}Z^4 = 3$  for  $Z \sim N(0, 1)$
- ▶ 95% interval for  $\text{IV}_t$ :  $\text{RV}_t \pm 1.96 \sqrt{\frac{2}{3} \sum_i r_{t,i}^4}$ ; it may contain negative values

- ▣ Log version (delta method): an interval for  $\ln \text{IV}_t$

$$\ln \text{RV}_t \pm 1.96 \frac{\sqrt{\frac{2}{3} \sum_i r_{t,i}^4}}{\text{RV}_t}$$

- ▶ exponentiated, it never contains negative values and has better coverage in finite samples

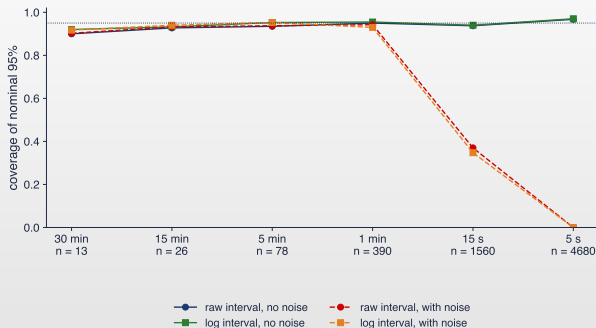
## A simulated market with a known truth

- One-second grid,  $n = 23\,400$  seconds a day; the log spot variance is the sum of two Gaussian AR(1) factors

$$\ln \sigma_s^2 = c + f_s^{(1)} + f_s^{(2)}$$

- ▶  $c$ : constant fixing the mean;  $f^{(1)}$ : slow factor (half-life 60 days, stationary s.d. 0.75);  $f^{(2)}$ : fast factor (half-life 2 days, s.d. 0.45)
- ▶ mean IV = 1 (%<sup>2</sup> a day), the mean of the S&P 500 realised kernel in 2000–2022; leverage: correlation  $-0.6$  between price shocks and shocks to the fast factor
- Jumps: Poisson with 0.08 jumps a day, sizes  $N(0, 0.8^2)$  (%): jumps make about 5.6% of quadratic variation
- Noise: we observe  $Y_s = X_s + u_s$ , with  $u_s$  i.i.d.  $N(0, \omega^2)$  and  $\omega = 0.004\%$ 
  - ▶  $Y_s$ : observed log price;  $X_s$ : efficient log price;  $u_s$ : microstructure noise with standard deviation  $\omega$ ; at one second,  $2n\omega^2 = 0.75$  (next slides)
- Each estimator below is compared with the true IV or QV of the same simulated day

## Coverage of the feasible CLT



- 400 simulated days without jumps; 95% intervals for  $IV_t$  (raw and log) from returns sampled every 30 minutes to every 5 seconds, without and with noise

## Interpreting the CLT coverage

- Without noise, coverage approaches 95% as  $n$  grows: raw interval 90.0% at  $n = 13$  and 95.0% at  $n = 390$ ; the log interval is closer at small  $n$  (92.0%)
- With noise, coverage collapses once sampling is too fine: 37.0% at 15 seconds and 0.0% at 5 seconds; at 1 minute it is still 94.2%
- The interval shrinks like  $n^{-1/2}$  while the noise bias grows like  $n$ : the CLT is a statement about the efficient price, not about the observed one
- Practical rule: 5-minute returns for liquid assets unless a noise-robust estimator is used

## Where the realised measures come from



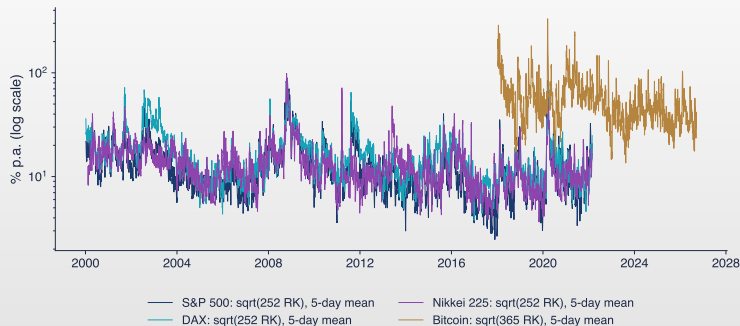
Ole E. Barndorff-Nielsen, 2007

Photo: Thomas Steiner (2007); CC BY-SA 2.5;

[Wikimedia Commons](#)

- [Heber, Lunde, Shephard and Sheppard \(2009\)](#), version 0.3: daily RV, BV, realised kernels and open-to-close returns for 31 indices, 2000 to 25 February 2022, built from cleaned tick data
- its terms: free use if the library and its version are cited; the archived copy keeps the 2022 file available after the shutdown
- Bitcoin and Ether trade 24/7 and Binance publishes all one-minute and one-second prices: we compute RV, BV, RQ, tripower quarticity and a realised kernel for each UTC day, 2018–2026
- No free source covers equity intraday data after 2022: a project can rebuild them from a licensed feed

## Realised volatility across markets



- Annualised realised kernel, 5-day means: S&P 500, DAX and Nikkei 225 (Oxford-Man, 2000–2022); Bitcoin (Binance one-minute prices, 2018–2026, 365 days a year)

## Interpreting the realised volatilities

- Average annualised volatility from the kernel: S&P 500 16.2%, DAX 19.5%, Nikkei 16.6%, Bitcoin 68.4%
- The equity series move together (2002, 2008, 2011, 2020): a common long-run factor, the multivariate theme of the last section
- Open-to-close measures miss the overnight return: for daily risk one adds the squared overnight return or scales the measure
- Bitcoin has no overnight gap, but its daily volatility is three to four times higher and falls after 2023

## Microstructure noise (1/2): the bias of RV

- The observed log price is the efficient price plus a noise term

$$Y_{i/n} = X_{i/n} + u_i$$

- ▶  $u_i$ : i.i.d. noise with variance  $\omega^2$ , independent of  $X$ ; sources: bid-ask bounce, price discreteness, stale quotes

- RV computed from the observed prices at  $n$  intervals,  $RV_t^{(n)}$ , is biased upwards, and the bias grows with  $n$   
( ▶ Appendix )

$$\mathbb{E}(RV_t^{(n)}|X) = IV_t + 2n\omega^2, \quad \text{Var}(RV_t^{(n)}|X) \approx 4n\mathbb{E}u^4$$

- ▶ each observed return contains  $u_i - u_{i-1}$ , with variance  $2\omega^2$ ; summed over  $n$  returns this gives  $2n\omega^2$
- ▶ RV diverges as  $n \rightarrow \infty$ : sampling as often as possible is not optimal

## Microstructure noise (2/2): diagnostics

- Two consequences of i.i.d. noise
  - ▶  $\hat{\omega}^2 = \text{RV}_t^{(n)} / (2n)$  at the highest frequency estimates the noise variance, since  $2n\omega^2$  dominates IV there
  - ▶ returns of the observed price have first-order autocovariance  $-\omega^2$ : negative autocorrelation is the signature of i.i.d. noise
- Signature plot: average  $\text{RV}^{(n)}$  against the sampling interval
  - ▶ a flat region shows the intervals where noise no longer matters
- Real noise is not i.i.d.: it is autocorrelated and correlated with the efficient price, especially in quote data  
Hansen and Lunde (2006b)

## Optimal sparse sampling

- Mean square error (MSE) of  $RV^{(n)}$  as an estimator of IV: sampling variance falls, noise bias rises with  $n$

$$MSE(n) \approx \underbrace{\frac{2IQ}{n}}_{\text{variance}} + \underbrace{(2n\omega^2)^2}_{\text{bias}^2} + 4n\mathbb{E}u^4 + \dots$$

- Minimising the first two terms gives the optimal number of intervals [Bandi and Russell \(2008\)](#)

$$n^* \approx \left( \frac{IQ}{4\omega^4} \right)^{1/3}$$

- ▶ the optimal grid is coarser when noise ( $\omega^2$ ) is large relative to volatility (IQ)
- ▶ example (Seminar 8, A3):  $IV = IQ = 1$ ,  $\omega^2 = 1.6 \times 10^{-5}$  gives  $n^* \approx 990$ , a return every 24 seconds
- The best sparse RV converges only at rate  $n^{1/6}$  and discards most data: this motivates estimators that use all observations
- Subsampling: average the RV of the  $K$  offset grids (starting at second  $0, 1, \dots, K-1$ ); same bias, smaller variance

## Two scales and realised kernels (1/2)

- Two-scales RV [Zhang, Mykland and Aït-Sahalia \(2005\)](#): the fast scale estimates the noise bias of the slow one and removes it

$$\text{TSRV} = \overline{\text{RV}}^{(K)} - \frac{\bar{n}}{n} \text{RV}^{(n)}, \quad \bar{n} = \frac{n - K + 1}{K}$$

- ▶  $\overline{\text{RV}}^{(K)}$ : subsampled RV on  $K$  offset sparse grids (slow scale), each with about  $\bar{n}$  returns;  $\text{RV}^{(n)}$ : RV on all  $n$  returns (fast scale)
- ▶ the bias of the slow scale is  $2\bar{n}\omega^2 = \frac{\bar{n}}{n} \cdot 2n\omega^2$ : the correction subtracts it; rate  $n^{1/6}$
- Pre-averaging [Jacod et al. \(2009\)](#): average returns over blocks before squaring
  - ▶ rate  $n^{1/4}$ , the optimal one in the presence of noise; in practice similar to the realised kernel

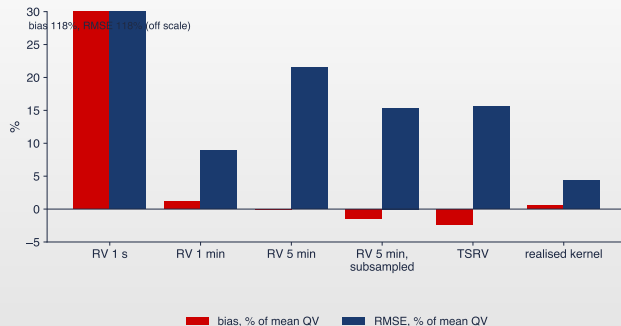
## Two scales and realised kernels (2/2): the realised kernel

- Realised kernel [Barndorff-Nielsen, Hansen, Lunde and Shephard \(2008\)](#): RV plus weighted realised autocovariances

$$RK_t = \gamma_0 + \sum_{h=1}^H k\left(\frac{h}{H+1}\right)(\gamma_h + \gamma_{-h}), \quad \gamma_h = \sum_i r_{t,i} r_{t,i-h}$$

- $\gamma_h$ : realised autocovariance at lag  $h$  ( $\gamma_0 = \text{RV}$ );  $H$ : bandwidth, the number of lags used;  $k(\cdot)$ : weight function with  $k(0) = 1$ ,  $k(1) = 0$
  - Parzen weights:  $k(x) = 1 - 6x^2 + 6x^3$  for  $x \leq \frac{1}{2}$  and  $2(1-x)^3$  for  $\frac{1}{2} < x \leq 1$
  - the negative  $\gamma_1$  created by noise cancels the bias  $2n\omega^2$ : the same idea as HAC variance estimation ([Chapter 0](#))
- Properties
  - nonnegative by construction; rate  $n^{1/5}$  with  $H = c^* \xi^{4/5} n^{3/5}$ , where  $\xi^2 = \omega^2 / \sqrt{IQ}$  is the noise-to-signal ratio and  $c^* = 3.51$  for Parzen [Barndorff-Nielsen et al. \(2009\)](#)
  - it also corrects serially dependent noise of short range, of either sign

## Estimators against the true quadratic variation

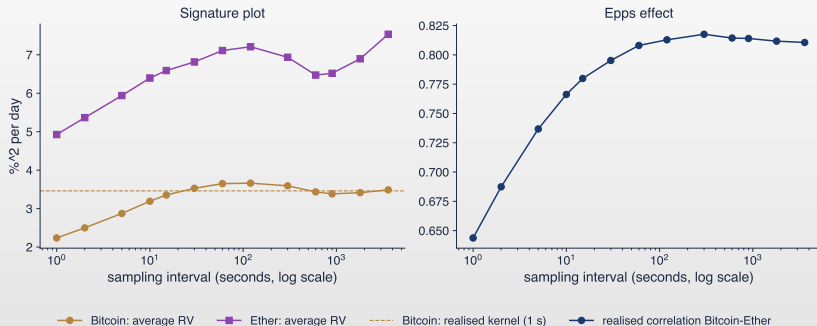


- 300 simulated days with jumps and noise; bias and root mean square error relative to the mean QV; TSRV with  $K = 300$  seconds; RK with the BNHLS bandwidth (median  $H = 31$ )

## Interpreting the estimator comparison

- ▣ RV at one second is dominated by noise: bias 117.8% of QV; at one minute the bias is 1.2% and the RMSE 8.9%
- ▣ Five-minute RV is unbiased but noisy (RMSE 21.6%); subsampling and TSRV reduce it to 15.4% and 15.6%
- ▣ The realised kernel uses all 23 400 returns: bias 0.5%, RMSE 4.4%, the best of the six
- ▣ All estimators here target QV, jumps included; separating the jump part needs the next section

## Signature plot and Epps effect: Bitcoin and Ether



- Binance one-second prices, August 2026 (31 days); average daily RV by sampling interval (subsampled); realised correlation of the two coins by interval

## Interpreting the signature plot

- Bitcoin: RV is 2.24 at one second, 3.65 at one minute, 3.59 at five minutes: the signature *rises*, the opposite of the i.i.d.-noise prediction
- Reason: 52% of one-second returns are zero (no trade or the same price) and prices adjust gradually, so one-second returns are positively autocorrelated
- The realised kernel on one-second data (3.46, median  $H = 22$ ) adds the positive autocovariances back and lands on the flat region of 1–30 minutes
- Epps effect [Epps \(1979\)](#): realised correlation 0.64 at one second against 0.82 at five minutes; asynchronous trading biases covariances towards zero at high frequency

## Recap: Realised measures

- ▣ RV estimates QV with error  $\sqrt{2IQ/n}$ ; the log interval behaves better; stable convergence makes studentization valid
- ▣ Noise biases RV at high frequency; its sign and size are empirical: look at the signature plot first
- ▣ Realised kernels use all the data and correct short-range dependent noise; 5-minute RV remains a robust benchmark

## Bipower variation (1/2)

- Barndorff-Nielsen and Shephard (2004): sum of products of adjacent absolute returns, rescaled

$$BV_t = \mu_1^{-2} \frac{n}{n-1} \sum_{i=2}^n |r_{t,i}| |r_{t,i-1}|, \quad \mu_1 = \mathbb{E}|Z| = \sqrt{2/\pi}$$

- ▶  $Z \sim N(0, 1)$ ;  $\mu_1^{-2}$  makes  $BV_t$  consistent for  $IV_t$ ;  $\frac{n}{n-1}$  corrects for the  $n-1$  products
- ▶  $BV_t \rightarrow IV_t$  even with (finite-activity) jumps: a jump enters only two products, each multiplied by a return of order  $n^{-1/2}$
- Hence  $RV_t - BV_t \rightarrow \sum_{s \leq 1} (\Delta J_s)^2$ : a nonparametric estimate of the jump variation of the day

## Bipower variation (2/2): related estimators

- ▣ Jump-robust quarticity: the tripower quarticity

$$\text{TQ}_t = n \mu_{4/3}^{-3} \frac{n}{n-2} \sum_{i=3}^n \prod_{j=0}^2 |r_{t,i-j}|^{4/3}, \quad \mu_{4/3} = \mathbb{E}|Z|^{4/3}$$

- ▶ products of three adjacent absolute returns, each to the power  $4/3$ ;  $\text{TQ}_t \rightarrow \text{IQ}_t$  even with jumps (RQ does not)
- ▣ Nearest-neighbour truncation (MedRV, MinRV) Andersen, Dobrev and Schaumburg (2012): smaller finite-sample bias from a jump and from zero returns
- ▣ Threshold (truncated) RV: drop the returns above  $c n^{-\varpi}$ 
  - ▶  $c > 0$ : a constant (a multiple of the local volatility);  $\varpi \in (0, \frac{1}{2})$ : the threshold shrinks more slowly than a diffusive return, so only jumps are removed

## Testing for jumps (1/2): the ratio statistic

- Barndorff-Nielsen and Shephard (2006): without jumps, the difference  $RV - BV$  is also mixed normal

$$\sqrt{n}(RV_t - BV_t) \rightarrow MN(0, \vartheta IQ_t), \quad \vartheta = \mu_1^{-4} + 2\mu_1^{-2} - 5 = \frac{\pi^2}{4} + \pi - 5 \approx 0.609$$

- ▶  $\vartheta$ : a constant that measures how much less efficient BV is than RV
- Ratio statistic Huang and Tauchen (2005): the relative jump contribution, studentised

$$z_t = \frac{(RV_t - BV_t)/RV_t}{\sqrt{\frac{\vartheta}{n} \max(1, TQ_t/BV_t^2)}} \rightarrow N(0, 1)$$

- ▶ one-sided test: reject “no jump on day  $t$ ” for large  $z_t$
- ▶ the ratio form and the max adjustment give the best size in their simulations;  $TQ_t/BV_t^2 \geq 1$  by Jensen when volatility varies within the day

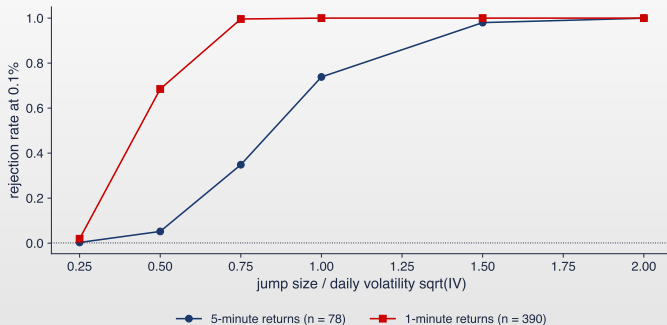
## Testing for jumps (2/2): in practice

- A daily test answers “was there a jump today?”; locating it within the day needs a test on each return standardised by local BV [Lee and Mykland \(2008\)](#)
- Multiple testing: at level  $\alpha$  over  $T$  days one expects  $\alpha T$  false jump days
  - ▶ use a small  $\alpha$  (0.1%) or a family-wise (FWER) or false-discovery-rate (FDR) correction ([Chapter 1](#))
- Separating the continuous and jump parts of RV [Andersen, Bollerslev and Diebold \(2007\)](#)

$$J_t = \mathbb{I}(z_t > z_{1-\alpha})(RV_t - BV_t), \quad C_t = RV_t - J_t$$

- ▶  $\mathbb{I}(\cdot)$ : indicator, 1 if the test rejects and 0 otherwise;  $z_{1-\alpha}$ : the  $1 - \alpha$  quantile of  $N(0, 1)$
- ▶  $J_t$ : jump part, nonzero only on significant days;  $C_t$ : continuous part

## Size and power of the ratio test

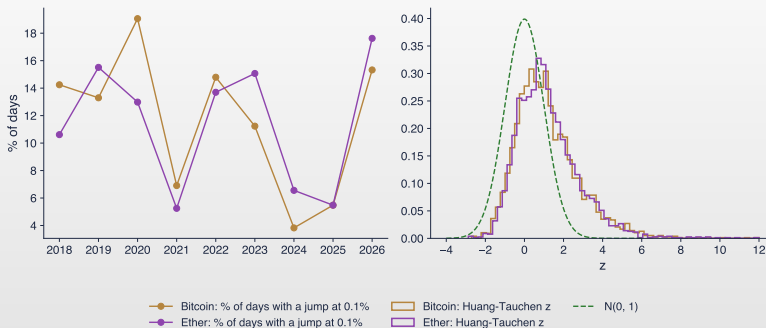


- 2000 simulated days (two-factor SV, no jumps), then one jump of size  $c\sqrt{IV_t}$  at a random time; one-sided test at 0.1% with 5-minute and 1-minute returns

## Interpreting the size and power

- Size without jumps: 0.0% (5 minutes) and 0.2% (1 minute) against the nominal 0.1%; with i.i.d. noise of the calibrated size still 0.1% at 1 minute
- Power for a jump of half a daily standard deviation: 5% with 5-minute returns, 68% with 1-minute returns; for one standard deviation 74% and 100%
- The power depends on  $c\sqrt{n}$ : small jumps are absorbed into the diffusion at coarse sampling (Seminar 8, A6 gives the detection threshold)
- The trade-off with noise is the same as for RV: finer sampling raises power until noise and zero returns distort the statistic

## Jump days of Bitcoin and Ether



- Huang–Tauchen test with 5-minute returns ( $n = 288$  a UTC day), 2018–2026,  $T = 3\,165$  days; left: share of jump days at 0.1% by year; right: distribution of  $z_t$  against  $N(0, 1)$

## Interpreting the crypto jump tests

- ▣ Bitcoin: 362 jump days (11.4%) where about 3.2 false rejections are expected; Ether: 355 days (11.2%)
- ▣ Yet the jumps are small: they make 3.0% (Bitcoin) and 2.3% (Ether) of total variation; the rest is continuous
- ▣ The whole distribution of  $z_t$  is shifted (mean 1.16, not 0): besides jumps, 24-hour intraday seasonality, zero returns and flash moves violate the assumptions of the null
- ▣ Before calling a day a jump day: standardise returns by an intraday volatility pattern, check zero returns, and control the number of tests

## Recap: Jumps

- BV estimates the continuous part;  $RV - BV$  the jump part; TQ makes the test robust to jumps in the quarticity
- The ratio test has good size in simulations; its power grows with  $c\sqrt{n}$
- In real data the null is fragile: many rejections, little jump variation; treat jump counts as model-dependent

## The HAR model of Corsi (1/2): the regression

- Corsi (2009): tomorrow's RV is a linear function of the daily, weekly and monthly averages of past RV

$$RV_{t+1} = \beta_0 + \beta_d RV_t + \beta_w RV_t^{(w)} + \beta_m RV_t^{(m)} + u_{t+1}$$

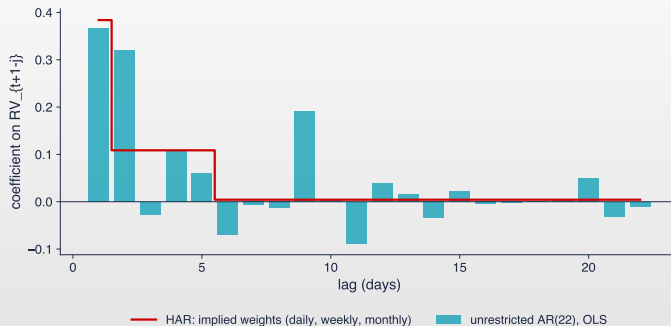
$$RV_t^{(w)} = \frac{1}{5} \sum_{j=0}^4 RV_{t-j}, \quad RV_t^{(m)} = \frac{1}{22} \sum_{j=0}^{21} RV_{t-j}$$

- ▶  $RV_t^{(w)}, RV_t^{(m)}$ : averages over the last 5 trading days (a week) and the last 22 (a month)
- ▶  $\beta_d, \beta_w, \beta_m \geq 0$ : weights of the three horizons;  $\beta_0$ : intercept;  $u_{t+1}$ : forecast error with mean zero
- ▶  $\beta_d + \beta_w + \beta_m$ : the persistence; the closer to 1, the slower volatility returns to its mean
- Heterogeneous market hypothesis: daily, weekly and monthly traders react to volatility at their own horizon
  - ▶ the cascade produces slowly decaying autocorrelations

## The HAR model of Corsi (2/2): estimation and variants

- ▣ Statistically, HAR is an AR(22) with 19 linear restrictions (step-shaped coefficients)
  - ▶ short memory, but it mimics long memory over a month
- ▣ OLS is consistent; the errors are heteroskedastic and, for multi-day targets, overlapping
  - ▶ use Newey–West [Newey and West \(1987\)](#) (HAC) standard errors
- ▣ Variants
  - ▶ log-HAR: the same regression on  $\ln RV$ , with near-Gaussian errors; the forecast of RV is  $\exp(\hat{\mu} + \frac{1}{2}\hat{\sigma}_u^2)$ , with  $\hat{\mu}$  the fitted log value and  $\hat{\sigma}_u^2$  the residual variance
  - ▶ HAR on  $\sqrt{RV}$ ;  $h$ -day targets (the average RV over the next  $h$  days)
- ▣ Extensions: continuous and jump parts, HAR-CJ [Andersen, Bollerslev and Diebold \(2007\)](#); positive and negative semivariances [Patton and Sheppard \(2015\)](#); implied volatility; leverage terms

## HAR as a restricted AR(22)



- S&P 500 (Oxford-Man, 5-minute RV, 2000–2022,  $T = 5\,530$ ): coefficients of an unrestricted AR(22) by OLS and the step weights implied by the HAR estimates

## Interpreting the HAR estimates

- ▣ HAR:  $\hat{\beta}_d = 0.275$  (0.104),  $\hat{\beta}_w = 0.522$  (0.140),  $\hat{\beta}_m = 0.094$  (0.078), Newey–West s.e.;  $R^2 = 0.560$ ; persistence  $\sum \hat{\beta} = 0.891$
- ▣ The AR(22) with 23 coefficients reaches  $R^2 = 0.596$  in sample, with erratic coefficients; HAR captures the shape with 4
- ▣ Log-HAR:  $\hat{\beta}_d = 0.393$ ,  $\hat{\beta}_w = 0.399$ ,  $\hat{\beta}_m = 0.153$ ,  $R^2 = 0.719$  on  $\ln RV$ ; in logs the monthly component matters more
- ▣ In levels a few crisis days dominate OLS: large standard errors on  $\beta_d$  and  $\beta_m$ ; weighted least squares or logs give more stable estimates

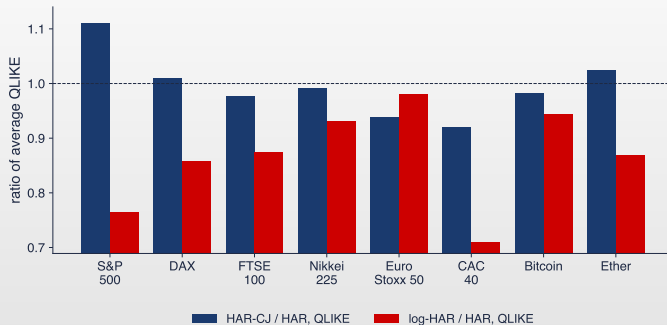
## Forecasting design

- ▣ One day ahead, rolling window of 1 000 days, re-estimated every day; target  $RV_{t+1}$
- ▣ Losses:  $MSE = (RV - f)^2$  and QLIKE, both robust to the noise in  $RV$  (see below)

$$QLIKE = \frac{RV}{f} - \ln \frac{RV}{f} - 1 \geq 0$$

- ▶  $f$ : the variance forecast; QLIKE is 0 when  $f = RV$  and penalises under-prediction more than over-prediction
- ▣ Insanity filter [Bollerslev, Patton and Quaadvlieg \(2016\)](#): a forecast outside the range of the in-sample  $RV$  is replaced by the in-sample mean
  - ▶ linear HAR-type forecasts can be negative after a spike; QLIKE is then undefined
  - ▶ the filter is part of the method and must be reported
- ▣ Diebold–Mariano [Diebold and Mariano \(1995\)](#)  $t$ -statistic of the loss differential against HAR
  - ▶ Newey–West variance; negative = better than HAR, below  $-1.96$  significant at 5%; several models: the model confidence set [Hansen, Lunde and Nason \(2011\)](#), Chapter 1
- ▣ Models: HAR, HAR-CJ (continuous part and jump part, from BV), log-HAR; for Bitcoin and Ether also HARQ

## HAR extensions out of sample



- Average QLIKE relative to HAR (below 1 = better), six Oxford-Man indices (2004–2022) and two cryptocurrencies (from 01.11.2020); 1 000-day rolling window

## Interpreting the out-of-sample HAR comparison

- Log-HAR beats HAR in QLIKE for 6 of 6 indices: S&P 500 0.764 (DM  $-2.07$ ), CAC 40 0.709 (DM  $-1.73$ ), FTSE 0.874 (DM  $-5.08$ )
- HAR-CJ helps in 4 of 6 indices, but not for the S&P 500 (1.110): separating jumps adds estimation noise when jumps are small
- The ranking depends on the loss: in MSE the gains of log-HAR are smaller (S&P 500: 0.837); MSE weighs the crisis days most
- Six indices from one data vendor and one period: report all of them, not the best one (data snooping, [Chapter 1](#))

## HARQ (1/2): the measurement error of RV

- RV is the latent IV plus a measurement error whose variance changes every day

$$RV_t = IV_t + e_t, \quad \text{Var}(e_t|IQ_t) \approx \frac{2 IQ_t}{n}$$

- ▶  $e_t$ : measurement error of day  $t$ ; on volatile days ( $IQ_t$  large) RV is less precise

- Errors in variables: in an AR(1) for IV with coefficient  $\phi$ , the OLS slope on RV is attenuated (▶ [Appendix](#))

$$\text{plim } \hat{\phi} = \phi\lambda, \quad \lambda = \frac{\text{Var}(IV)}{\text{Var}(IV) + \mathbb{E}\text{Var}(e)} \in (0, 1)$$

- ▶  $\lambda$ : reliability ratio, the share of the variance of RV that is signal
- ▶ one constant  $\beta_d$  is a compromise: too high on noisy days, too low on precise days

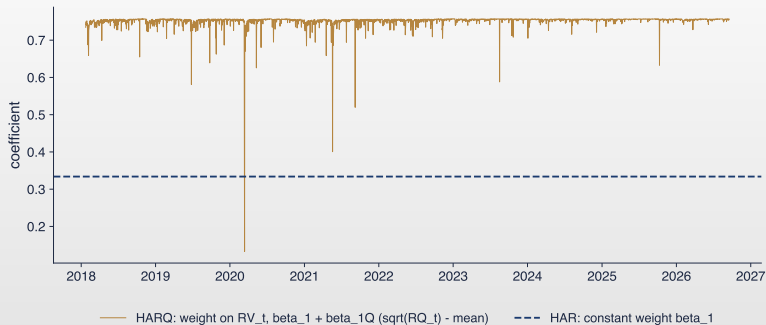
## HARQ (2/2): a weight that depends on precision

- Bollerslev, Patton and Quaadvlieg (2016): the weight of yesterday's RV moves with its estimated precision

$$RV_{t+1} = \beta_0 + (\beta_d + \beta_{dQ} \sqrt{RQ_t}) RV_t + \beta_w RV_t^{(w)} + \beta_m RV_t^{(m)} + u_{t+1}$$

- ▶  $RQ_t$ : realised quarticity, the estimate of  $IQ_t$ ;  $\sqrt{RQ_t}$  is proportional to the standard deviation of  $e_t$
- ▶  $\beta_{dQ} < 0$  expected: the less precise yesterday's RV, the lower its weight
- ▶  $\sqrt{RQ_t}$  is demeaned, so  $\beta_d$  is the weight on a day of average precision
- One extra parameter, still OLS
- The original paper: S&P 500 futures and 27 Dow Jones stocks, gains over HAR in and out of sample
  - ▶ our replication extends it to Bitcoin and Ether, 2018–2026, where RQ is available from our one-minute data

## HARQ for Bitcoin: a weight that moves with precision



- Full-sample HARQ on Bitcoin 5-minute RV and RQ: daily weight  $\hat{\beta}_d + \hat{\beta}_{dQ}(\sqrt{RQ_t} - \sqrt{RQ})$  against the constant HAR weight

## Interpreting HARQ

- ▣  $\hat{\beta}_{dQ} = -0.00021$  ( $t = -2.7$ , Newey–West): negative, as predicted; the daily weight at average precision is 0.752, against 0.334 in HAR
- ▣ On most days the weight is near 0.76; on the noisiest days it falls (1% quantile 0.71, minimum 0.13): yesterday's RV is trusted only when it is precise
- ▣ Out of sample, QLIKE relative to HAR: Bitcoin 0.946 (DM  $-2.44$ ), Ether 0.921 (DM  $-4.50$ ): the gain of the original study carries over to crypto
- ▣ In MSE the picture differs (Bitcoin 1.163): MSE is dominated by a few extreme days, where HARQ shrinks the most; the AI mini-case checks how fragile the gain is

## Recap: HAR and HARQ

- ▣ HAR is a parsimonious restricted AR(22) estimated by OLS; logs stabilise it
- ▣ Out-of-sample gains are loss- and market-specific: report all series, the filter and DM tests
- ▣ HARQ lets the measurement error of RV set the weight of yesterday: one parameter, a robust gain in QLIKE

## Realized GARCH (1/2): the three equations

- Hansen, Huang and Shek (2012), log-linear form: a return equation, a GARCH equation driven by the realised measure, and a measurement equation

$$\begin{aligned} r_t &= \sqrt{h_t} z_t, & \ln h_t &= \omega + \beta \ln h_{t-1} + \gamma \ln x_{t-1} \\ & & \ln x_t &= \xi + \varphi \ln h_t + \tau(z_t) + u_t \end{aligned}$$

- ▶  $h_t$ : conditional variance of the return  $r_t$ ;  $z_t$ : i.i.d. standardised shock;  $x_t$ : realised measure of day  $t$  (here the realised kernel RK)
- ▶  $\beta$ : persistence of  $\ln h_t$ ;  $\gamma$ : weight of yesterday's realised measure (the news variable that replaces  $\varepsilon_{t-1}^2$ )
- ▶  $u_t$ : measurement error, i.i.d.  $N(0, \sigma_u^2)$ , independent of  $z_t$
- Measurement equation:  $x_t$  is a noisy, possibly biased signal of  $h_t$ 
  - ▶  $\varphi = 1$ :  $x_t$  proportional to  $h_t$ ;  $\xi$  absorbs the scale (open-to-close measure against close-to-close variance)
  - ▶ leverage function  $\tau(z) = \tau_1 z + \tau_2(z^2 - 1)$ : with  $\tau_1 < 0$ , negative returns raise the next realised measure (news impact, TSA, Chapter 5)

## Realized GARCH (2/2): reduced form and estimation

- ▣ Substituting the measurement equation into the GARCH equation:  $\ln h_t$  is an AR(1)

$$\ln h_t = (\omega + \gamma\xi) + \pi \ln h_{t-1} + \gamma(\tau(z_{t-1}) + u_{t-1}), \quad \pi = \beta + \varphi\gamma$$

- ▶  $\pi$ : persistence of the log variance; multi-step forecasts follow, because  $x_t$  has its own equation
- ▣ Joint QML: the log-likelihood sums a return part and a measurement part

$$\ell = -\frac{1}{2} \sum_t \left[ \underbrace{\ln h_t + z_t^2}_{r_t} + \underbrace{\ln \sigma_u^2 + u_t^2 / \sigma_u^2}_{x_t} \right]$$

- ▶ comparison with GARCH only through the partial likelihood of  $r_t$  (the first part), since GARCH does not model  $x_t$

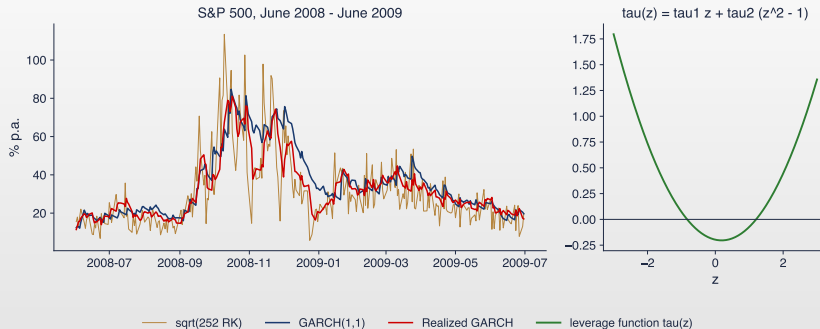
## HEAVY and the family of realised-measure models

- HEAVY Shephard and Sheppard (2010): two GARCH-type equations, one for the return variance and one for the mean of the realised measure

$$\begin{aligned}h_t &= \text{Var}(r_t | \mathcal{F}_{t-1}) = \omega + \alpha \text{RM}_{t-1} + \beta h_{t-1} \\ \mu_t &= \mathbb{E}(\text{RM}_t | \mathcal{F}_{t-1}) = \omega_R + \alpha_R \text{RM}_{t-1} + \beta_R \mu_{t-1}\end{aligned}$$

- ▶  $\text{RM}_t$ : realised measure of day  $t$ ;  $(\omega, \alpha, \beta)$  and  $(\omega_R, \alpha_R, \beta_R)$ : the parameters of the two equations
- ▶ each equation is estimated separately by QML; momentum: after a shock,  $h_t$  keeps rising while  $\mu_t$  catches up
- ▶ when the realised measure is in the variance equation, the squared return usually adds nothing ( $\varepsilon_{t-1}^2$  gets a zero coefficient)
- Realized EGARCH Hansen and Huang (2016): several realised measures and a richer leverage; score-driven versions exist for fat tails
- Compared with HAR: returns and realised measures are modelled jointly, so densities and VaR/ES forecasts are available (Chapter 9)

## Case study: Realized GARCH for the S&P 500



- Open-to-close returns and the Parzen realised kernel (Oxford-Man), 2000–2022,  $T = 5552$ , as in Hansen, Huang and Shek (2012); left: the 2008 crisis; right: the estimated leverage function

## Realized GARCH estimates

| $\omega$         | $\beta$          | $\gamma$         | $\xi$             | $\varphi$        | $\tau_1$          | $\tau_2$         | $\sigma_u$       |
|------------------|------------------|------------------|-------------------|------------------|-------------------|------------------|------------------|
| 0.119<br>(0.010) | 0.678<br>(0.018) | 0.287<br>(0.016) | -0.456<br>(0.021) | 1.012<br>(0.026) | -0.073<br>(0.008) | 0.197<br>(0.010) | 0.671<br>(0.008) |

- Bollerslev–Wooldridge s.e. in brackets; persistence  $\hat{\pi} = \hat{\beta} + \hat{\varphi}\hat{\gamma} = 0.968$  (GARCH(1,1) on the same returns: 0.989)
- Partial log-likelihood of the returns: Realized GARCH -6899.7, HEAVY -6902.8, GJR -6998.8, GARCH(1,1) -7099.0
- HEAVY variance equation:  $\hat{\alpha} = 0.377$  on  $RK_{t-1}$ ,  $\hat{\beta} = 0.691$

## Interpreting Realized GARCH

- ▣  $\hat{\phi} = 1.012$ : the kernel is proportional to the conditional variance;  $\hat{\xi} = -0.456$  with  $\hat{\sigma}_u = 0.671$ : on average the kernel lies below the conditional variance of the open-to-close return
- ▣  $\hat{\gamma} = 0.287$  is large and  $\hat{\beta} = 0.678$  smaller than in GARCH: the realised measure brings information faster than squared returns (left panel, autumn 2008)
- ▣ Leverage:  $\hat{\tau}_1 = -0.073$ ,  $\hat{\tau}_2 = 0.197$ : an asymmetric parabola, larger for negative  $z$
- ▣ The gain in the partial likelihood over GARCH(1,1) (199.3 points) has the direction of the original paper; HEAVY is almost as good with three parameters

## Recap: Realised-measure GARCH

- ▣ A realised measure in the variance equation beats squared returns as the news variable
- ▣ Realized GARCH adds a measurement equation: multi-step forecasts, leverage and a joint likelihood
- ▣ Compare with GARCH only on the partial likelihood of returns

## The proxy problem and robust losses (1/2)

- We never see  $\sigma_t^2$ : a forecast  $h_t$  is compared with a proxy  $\hat{\sigma}_t^2$  (squared return, RV, RK)
  - ▶ the proxy is conditionally unbiased,  $\mathbb{E}(\hat{\sigma}_t^2 | \mathcal{F}_{t-1}) = \sigma_t^2$ , but noisy
- A loss  $L$  is *robust* [Patton \(2011\)](#) if the ranking of forecasts by  $\mathbb{E}L(\hat{\sigma}_t^2, h_t)$  equals the ranking by  $\mathbb{E}L(\sigma_t^2, h_t)$ , for every unbiased proxy [Hansen and Lunde \(2006a\)](#)
- Necessary and sufficient condition: a Bregman form ([Chapter 1](#), consistent scoring functions for the mean)

$$L(\hat{\sigma}^2, h) = \tilde{C}(h) + B(\hat{\sigma}^2) + C(h)(\hat{\sigma}^2 - h), \quad C'(h) < 0$$

- ▶  $C$ : a decreasing function of the forecast;  $\tilde{C}$ : its antiderivative,  $\tilde{C}' = C$ ;  $B$ : any function of the proxy alone (it does not affect the ranking)

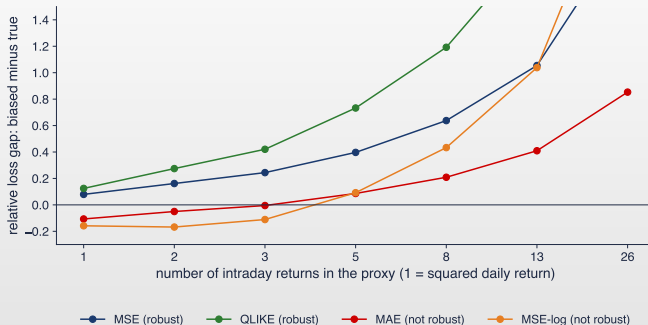
## The proxy problem and robust losses (2/2)

- ▣ Robust: MSE and QLIKE, the members of degree 2 and degree 0 of the homogeneous robust family

$$\text{MSE} = (\hat{\sigma}^2 - h)^2, \quad \text{QLIKE} = \frac{\hat{\sigma}^2}{h} - \ln \frac{\hat{\sigma}^2}{h} - 1$$

- ▶ degree: how the loss scales when proxy and forecast are multiplied by the same constant; QLIKE (degree 0) is scale-free and penalises under-prediction more
- ▣ Not robust: MAE =  $|\hat{\sigma}^2 - h|$ , MSE on logs ( $\ln \hat{\sigma}^2 - \ln h$ )<sup>2</sup>, MSE on standard deviations
  - ▶ with a noisy proxy they reward forecasts that are biased downwards
- ▣ Example: with  $\hat{\sigma}^2 = r^2$ , the MSE-log optimum is  $h^* = \exp(\mathbb{E} \ln r^2) = 0.28 \sigma^2$ 
  - ▶ because  $r^2 = \sigma^2 \chi_1^2$  for a Gaussian return and  $\mathbb{E} \ln \chi_1^2 = -1.27$ : MSE-log prefers a forecast far below the truth

## Robust and non-robust losses

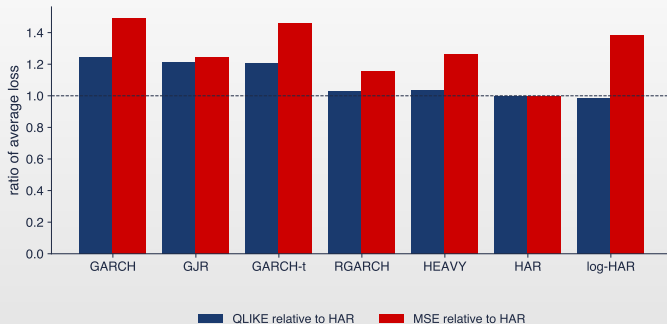


- True variance against a forecast biased down by the factor 0.6; proxy = RV from  $n$  intraday returns ( $n = 1$ : squared daily return); relative gap of expected losses, above zero = the true variance wins

## Interpreting the robustness experiment

- ▣ With the squared return as proxy, MAE prefers the biased forecast (gap  $-10.7\%$ ) and so does MSE-log ( $-15.8\%$ ); MSE ( $7.9\%$ ) and QLIKE ( $12.5\%$ ) do not
- ▣ With  $n = 5$  intraday returns all four losses rank correctly: a precise proxy makes non-robust losses “almost robust efHLb
- ▣ The danger is largest exactly where realised measures are unavailable: daily data, emerging markets, long histories
- ▣ Rule: evaluate volatility forecasts with QLIKE (and MSE), against the best available proxy

## Seven forecasts of S&P 500 volatility



- Open-to-close variance, one day ahead, 2016–2022 (1 537 days); expanding window from 2000, parameters re-estimated every 250 days; proxy: realised kernel

## Interpreting the forecast comparison

- ▣ QLIKE: HAR 0.328, log-HAR 0.324, Realized GARCH 0.337, HEAVY 0.339, GARCH- $t$  0.396, GJR 0.398, GARCH 0.407
- ▣ DM against HAR (QLIKE): GARCH 8.62, GJR 6.27, Realized GARCH 1.42, log-HAR  $-0.37$ : realised-measure models beat return-only models clearly; among them the differences are small
- ▣ In MSE, HAR (1.60) leads and the DM statistics are smaller: the 2020 crash dominates the squared errors
- ▣ One proxy (RK) for all models: the HAR family is fitted to that same measure, which favours it; a fair comparison also tries another proxy (RV, BV)

## Recap: Evaluation

- Only losses of the Patton class keep the ranking under a noisy unbiased proxy: MSE and QLIKE
- Realised-measure models beat return-only GARCH one day ahead; DM and the MCS of [Chapter 1](#) quantify by how much
- State the proxy, the loss, the window and the filter: each can change the ranking

## From one variance to a covariance matrix (1/2)

- The vector of  $N$  return shocks is a matrix square root of the conditional covariance times a standardised vector

$$\varepsilon_t = H_t^{1/2} \eta_t, \quad \eta_t \text{ i.i.d. } (0, I_N)$$

- ▶  $H_t = \text{Var}(\varepsilon_t | \mathcal{F}_{t-1})$ :  $N \times N$  conditional covariance matrix;  $I_N$ : identity matrix
- ▶  $H_t$  must be symmetric positive definite for every  $t$  and every parameter value
- VEC: every element of  $H_t$  depends on all past squares and cross-products

$$\text{vech}(H_t) = c + A \text{vech}(\varepsilon_{t-1} \varepsilon_{t-1}') + B \text{vech}(H_{t-1})$$

- ▶  $\text{vech}$ : stacks the  $N(N+1)/2$  distinct elements of a symmetric matrix;  $c$ : vector,  $A, B$ : square matrices of that size
- ▶ general, but positivity is hard to impose

## From one variance to a covariance matrix (2/2): BEKK, CCC and DCC

- BEKK Engle and Kroner (1995): quadratic forms guarantee positive definiteness

$$H_t = CC' + A'\varepsilon_{t-1}\varepsilon_{t-1}'A + B'H_{t-1}B$$

- ▶  $C$ : lower-triangular  $N \times N$ ;  $A, B$ :  $N \times N$  matrices; diagonal and scalar versions restrict  $A, B$
- ▶ identification:  $(A, B)$  and  $(-A, -B)$  give the same  $H_t$ ; fix the sign of one element

- CCC Bollerslev (1990) and DCC Engle (2002): variances from univariate GARCH, correlations modelled separately

$$H_t = D_t R D_t \quad (\text{CCC}), \quad H_t = D_t R_t D_t \quad (\text{DCC}), \quad D_t = \text{diag}(\sigma_{1t}, \dots, \sigma_{Nt})$$

- ▶  $\sigma_{it}$ : conditional standard deviation of asset  $i$  from its own GARCH;  $R$ : constant correlation matrix;  $R_t$ : time-varying correlation matrix

## The curse of dimensionality in numbers

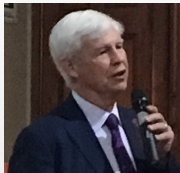
| $N$ | VEC        | BEKK   | diagonal BEKK | scalar BEKK | DCC   |
|-----|------------|--------|---------------|-------------|-------|
| 2   | 21         | 11     | 7             | 5           | 9     |
| 5   | 465        | 65     | 25            | 17          | 27    |
| 10  | 6 105      | 255    | 75            | 57          | 77    |
| 25  | 211 575    | 1 575  | 375           | 327         | 377   |
| 50  | 3 252 525  | 6 275  | 1 375         | 1 277       | 1 377 |
| 100 | 51 010 050 | 25 050 | 5 250         | 5 052       | 5 252 |

- Counts of free parameters of the (1,1) versions; DCC:  $3N$  univariate GARCH parameters, 2 correlation parameters and the  $N(N-1)/2$  elements of the target
- The DCC target is estimated by moments (correlation targeting), so the numerical search has only 2 parameters at any  $N$ ; the price:  $N(N-1)/2$  estimated correlations

## Asymptotic theory for multivariate GARCH

- Gaussian QML is the standard estimator; consistency and asymptotic normality: VEC and BEKK [Comte and Lieberman \(2003\)](#), [Hafner and Preminger \(2009\)](#); CCC with ARMA means [Ling and McAleer \(2003\)](#)
- Conditions are stronger than in the univariate case (moments of order 6 or 8 for BEKK in early results); the sandwich of the univariate section carries over
- DCC is estimated in two steps [Engle and Sheppard \(2001\)](#): (1) univariate GARCH for each series; (2) QML of the correlation part given the standardised residuals  $z_t = D_t^{-1} \varepsilon_t$ 
  - ▶ second-step standard errors must account for the first step (a GMM-type correction); naive second-step s.e. are too small
- Large  $N$ : the full likelihood needs  $R_t^{-1}$  and  $|R_t|$  at each  $t$  ( $O(N^3)$ ) and the score is dominated by noise; composite likelihood over pairs [Pakel, Shephard, Sheppard and Engle \(2021\)](#) avoids both

## DCC and its corrected version cDCC (1/2): DCC



Robert F. Engle, 2017

Photo: Econterms (2017); CC BY-SA 4.0;

[Wikimedia Commons](#)

- DCC: a GARCH(1,1)-type recursion for a quasi-correlation matrix  $Q_t$ , rescaled into a correlation matrix

$$Q_t = (1 - a - b)S + a z_{t-1} z'_{t-1} + b Q_{t-1}$$

$$R_t = \text{diag}(Q_t)^{-1/2} Q_t \text{diag}(Q_t)^{-1/2}$$

- ▶  $z_t = D_t^{-1} \varepsilon_t$ : standardised residuals of the univariate GARCH models;  $S$ : their sample correlation matrix (the target)
- ▶  $a \geq 0$ : reaction of correlations to yesterday's co-movement;  $b \geq 0$ : persistence;  $a + b < 1$
- ▶  $\text{diag}(Q_t)$ : the diagonal matrix of the diagonal elements of  $Q_t$ ; the rescaling puts ones on the diagonal of  $R_t$

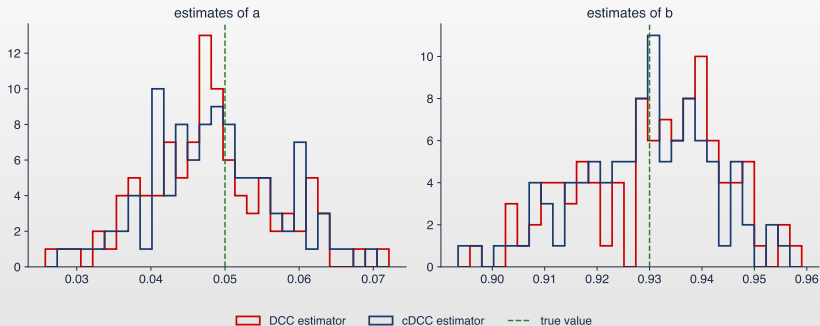
## DCC and its corrected version cDCC (2/2): cDCC

- Aielli (2013): the DCC target is estimated inconsistently
  - ▶  $\mathbb{E}(z_t z_t' | \mathcal{F}_{t-1}) = R_t \neq Q_t$ , so in general  $\mathbb{E}Q_t \neq \mathbb{E}z_t z_t'$
  - ▶ the moment estimator of  $S$  is inconsistent, and so is  $(\hat{a}, \hat{b})$
- cDCC: rescale the residuals by the diagonal of  $Q_{t-1}$  before they enter the recursion

$$z_{t-1}^* = \text{diag}(Q_{t-1})^{1/2} z_{t-1}, \quad Q_t = (1 - a - b)S + a z_{t-1}^* z_{t-1}^{*'} + b Q_{t-1}$$

- ▶ then  $\mathbb{E}(z_t^* z_t^{*'} | \mathcal{F}_{t-1}) = Q_t$ , and  $S = \mathbb{E}z_t^* z_t^{*'}$  is a valid target
- ▶ the cDCC target depends on  $(a, b)$ : it is estimated iteratively inside the likelihood

## DCC against cDCC: a simulation



- Bivariate cDCC process with  $a = 0.05$ ,  $b = 0.93$ , target correlation 0.5, unit variances;  $T = 2\,000$ , 100 replications; both estimators on each sample

## Interpreting the DCC simulation

- Mean estimates: DCC  $\hat{a} = 0.0478$ ,  $\hat{b} = 0.9307$ ; cDCC  $\hat{a} = 0.0486$ ,  $\hat{b} = 0.9288$  (true 0.05 and 0.93)
- RMSE of  $\hat{a}$ : 0.0090 (DCC) and 0.0086 (cDCC); of  $\hat{b}$ : 0.0135 and 0.0129: the inconsistency is real but small at these persistence levels
- Sampling error dominates the bias with two thousand observations; the difference matters more for the target in large systems and for inference on  $(a, b)$
- A proof of inconsistency does not tell you the size of the bias: simulate it in your own design before choosing an estimator

## Large covariance matrices: shrinkage and DCC-NL (1/2)

- With  $N/T$  not small, the eigenvalues of a sample covariance  $S$  are too dispersed
  - ▶  $N$ : number of assets;  $T$ : number of observations; the largest eigenvalues are too large, the smallest too small; the inverse  $S^{-1}$  amplifies the error
- Global minimum-variance (GMV) portfolio: the weights with the smallest variance that sum to 1

$$w = \frac{\Sigma^{-1} \mathbf{1}}{\mathbf{1}' \Sigma^{-1} \mathbf{1}}$$

- ▶  $\Sigma$ : covariance matrix of the returns;  $\mathbf{1}$ : vector of ones
- ▶ it loads on the directions with the smallest, most underestimated eigenvalues
- ▶ it depends only on  $\Sigma$ , so its realised risk measures the quality of the covariance forecast without noise from expected returns

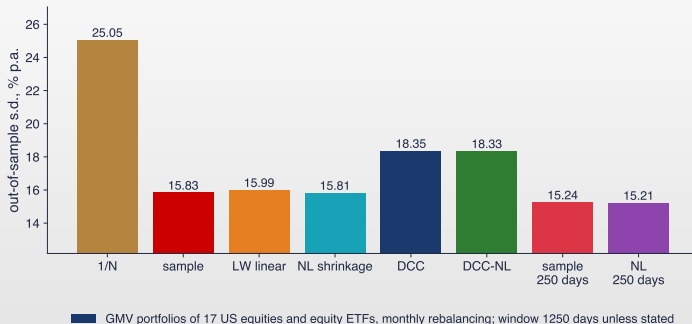
## Large covariance matrices: shrinkage and DCC-NL (2/2)

- Linear shrinkage [Ledoit and Wolf \(2004\)](#): a weighted average of  $S$  and a scaled identity

$$\hat{\Sigma} = \delta \mu I + (1 - \delta) S$$

- ▶  $\mu$ : average eigenvalue of  $S$ ;  $\delta \in [0, 1]$ : shrinkage intensity, estimated from the data (larger when  $N/T$  is larger)
- Nonlinear shrinkage [Ledoit and Wolf \(2020\)](#): keep the eigenvectors of  $S$ , replace each eigenvalue  $\lambda_i$  by  $d(\lambda_i)$ 
  - ▶  $d(\cdot)$ : an analytical kernel formula that pulls small eigenvalues up and large ones down, by different amounts
- DCC-NL [Engle, Ledoit and Wolf \(2019\)](#): in DCC, the target  $S$  of the standardised residuals is itself a large sample covariance: replace it by its nonlinear shrinkage
  - ▶ composite likelihood for  $(a, b)$ , univariate GARCH for the variances; evaluated by the out-of-sample s.d. of GMV portfolios

## Case study: Engle, Ledoit and Wolf (2019) on US equities



- 17 assets: 14 US stocks and three equity ETFs (SPY, QQQ, RSP), which are portfolios of stocks; GMV weights rebalanced every 21 days from 23.05.2017 (112 rebalancings); window 1 250 days, and 250 days for the last two bars; annualised out-of-sample s.d.

## Interpreting the GMV comparison

- Out-of-sample s.d. (% p.a.):  $1/N$  25.05, sample 15.83, linear shrinkage 15.99, nonlinear shrinkage 15.81, DCC 18.35, DCC-NL 18.33
- With  $N/T = 17/1250$  the sample covariance is already accurate: nonlinear shrinkage changes little (15.81 against 15.83); DCC-NL is no better than DCC
- Here the dynamics hurt: DCC (with GARCH variances, forecasts averaged over the 21-day holding period) gives 18.35 against 15.83; one-month-ahead variance forecasts of single stocks such as MSTR and TSLA are noisy, and GMV amplifies their errors
- With a window of 250 days ( $N/T$  five times larger) the sample GMV has s.d. 15.24 and nonlinear shrinkage 15.21: only 0.2% lower, although the ETFs make the correlation matrix nearly singular (median condition number 1146); the shorter window itself lowers the risk, a crude form of time variation
- The original study uses hundreds of stocks, where  $N/T$  is large and DCC-NL gains are substantial; with our  $N$  the lesson is when shrinkage matters, not how much it gains

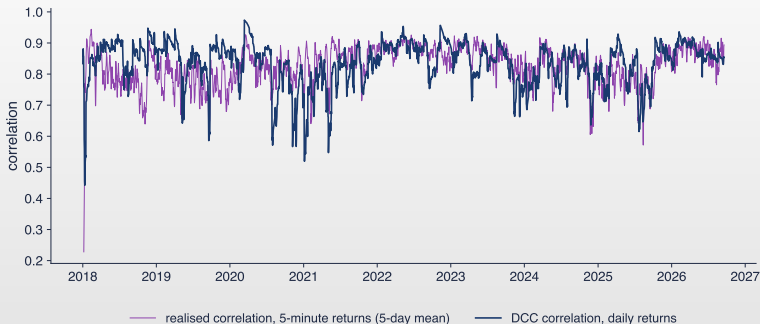
## Realised covariance

- The multivariate RV: the sum of outer products of the intraday return vectors

$$\text{RCov}_t = \sum_{i=1}^n r_{t,i} r'_{t,i} \rightarrow \int_0^1 \Sigma_s ds$$

- ▶  $r_{t,i}$ :  $N \times 1$  vector of intraday returns;  $\Sigma_s$ : spot covariance matrix; the limit adds the co-jumps (simultaneous jumps); with synchronous, noise-free prices it inherits the CLT of RV
- Asynchronous trading: previous-tick prices create zero returns for the asset that has not traded; covariances shrink towards zero as the interval falls (the Epps effect)
  - ▶ refresh-time sampling (all assets have traded) and the multivariate realised kernel [Barndorff-Nielsen et al. \(2011\)](#): consistent and positive semi-definite
- Forecasting: HAR on the elements of the Cholesky factor of  $\text{RCov}_t$  keeps forecasts positive definite [Chiriac and Voev \(2011\)](#); HEAVY and Realized GARCH have multivariate versions
- Evaluating covariance forecasts  $H_t$  against a matrix proxy  $\Sigma_t$ : robust losses [Laurent, Rombouts and Violante \(2012\)](#)
  - ▶ matrix QLIKE  $\ln |H_t| + \text{tr}(H_t^{-1} \Sigma_t)$ , with  $|\cdot|$  the determinant and  $\text{tr}$  the trace; Frobenius MSE, the sum of squared element-wise errors

## Bitcoin and Ether: realised and DCC correlation



- Daily realised correlation from 5-minute returns (5-day means) and the DCC correlation of daily close-to-close (UTC) returns, 2018–2026

## Interpreting the correlations

- Average realised correlation 0.83, average DCC correlation 0.83; correlation between the two series 0.26
- DCC uses one observation a day ( $\hat{a} = 0.051$ ,  $\hat{b} = 0.928$ ): it smooths and reacts with a lag; the realised measure moves daily and drops quickly in idiosyncratic episodes (5% quantile 0.68)
- Combining both (a realised measure in the correlation equation) is the multivariate analogue of Realized GARCH
- Five-minute sampling sits on the flat part of the Epps curve for these two coins (signature slide): the realised correlation is not biased towards zero

## Recap: Multivariate

- BEKK guarantees positivity at a quadratic parameter cost; DCC separates variances and correlations and scales to large  $N$
- cDCC fixes the inconsistency of the DCC target; in moderate designs the difference is small
- In high dimension, the target needs shrinkage (DCC-NL); evaluate with GMV risk or robust matrix losses

## An open question

- Does exploiting the measurement error of realised variance (HARQ) improve volatility forecasts robustly, or only under particular design choices?
  - ▶ formal: the QLIKE ratio  $\text{HARQ}/\text{HAR}$  and its DM statistic across a pre-registered grid (window, filter, realised measure, asset); falsified if the ratio exceeds 1 or the DM test is insignificant for most cells
  - ▶ the measure of precision (RQ) is itself noisy and fat-tailed, so the gain may be fragile
- Why it matters: risk systems use one-day volatility forecasts every day; a small but robust gain is worth more than a large fragile one
  - ▶ literature to start from: [Bollerslev, Patton and Quaedvlieg \(2016\)](#), [Corsi \(2009\)](#), [Patton \(2011\)](#), [Hansen and Lunde \(2006a\)](#), [Hansen, Lunde and Nason \(2011\)](#)

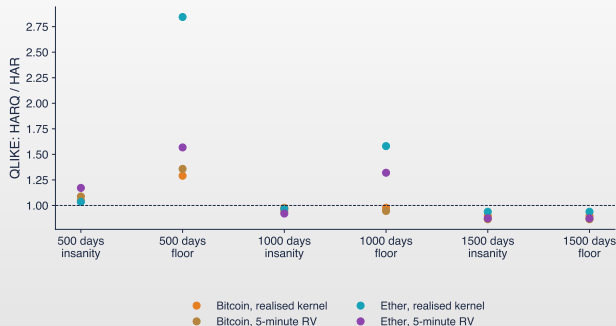
## The discovery loop with an AI assistant

- An AI assistant (an LLM such as Claude, ChatGPT, Gemini or Copilot) speeds up each step; Semantic Scholar and Elicit help with the literature
  - ▶ **literature:** List peer-reviewed papers since 2016 that test HARQ-type models out of sample on assets other than US equities; give DOIs. Then check every DOI on Crossref
  - ▶ **hypothesis:** Under which data features (24/7 trading, jumps, zero returns) should the HARQ gain shrink or grow? Give testable predictions.
  - ▶ **code and replication:** ask for HAR and HARQ rolling forecasts, then reproduce a known number first (the Bitcoin QLIKE ratio of this lecture)
  - ▶ **robustness and critique:** Act as a hostile referee: which choices in a HARQ forecast comparison could produce a spurious gain?
- Report: what was asked, what was kept, what was rejected (AI\_USE.md, AI\_ERRORS.md)

## What the human checks

- ▣ Every reference exists and says what is claimed (DOI resolves, title matches, the result is in the paper)
- ▣ The forecast at  $t$  uses only data up to  $t$ : windows, demeaning of RQ, the insanity filter and the bias correction of log forecasts are all computed in the estimation window
- ▣ The loss is robust (QLIKE, MSE) and the proxy is stated; DM statistics use HAC variances
- ▣ All cells of the grid are reported, not the best one; the number of comparisons is stated
- ▣ Realised measures are recomputed from raw prices and agree with an independent source on overlapping assets and dates

## Mini-case: twelve choices, one conclusion?



- QLIKE ratio HARQ/HAR for Bitcoin and Ether: window 500, 1 000 or 1 500 days; insanity filter or a floor at the smallest in-sample RV; target 5-minute RV or realised kernel
- Ratios from 0.865 to 2.844; HARQ better in 58% of the 24 cells, significantly better ( $DM < -1.96$ ) in 50%, significantly worse in 12%: an AI summary reporting one cell would hide most of this

## Project idea

- ▣ **Realised volatility of crypto-assets: does the equity toolkit transfer?:** replicate first, then extend
  - ▶ replicate: HAR and HARQ of this lecture for Bitcoin (QLIKE ratio 0.946) and Realized GARCH for the S&P 500 (persistence 0.968)
  - ▶ extend: ten coins from the Binance archive; noise-robust measures from one-second data; an intraday seasonality correction before jump tests; Realized GARCH with the realised kernel; the spot-ETF date (January 2024) as a natural break
  - ▶ pre-register: assets, sample, measures, models, windows, losses, the DM/MCS tests and the robustness grid
- ▣ Deliverables follow the course rules: repository, report, AI\_USE.md, AI\_ERRORS.md, oral defence

## Key takeaways

- ▣ Gaussian QML needs only the variance equation; with fat tails, use sandwich standard errors
- ▣ Volatility has fast and slow components; macro drivers of the slow one need long samples and honest variance ratios
- ▣ Realised measures make volatility observable with a known error; noise and jumps decide the estimator
- ▣ HAR, HARQ and Realized GARCH beat return-only models one day ahead; evaluate with QLIKE or MSE
- ▣ In many dimensions the correlation target needs shrinkage (DCC-NL); whether the dynamics help must be checked out of sample

## Self-assessment

### Questions

- ▣ Why are Hessian standard errors too small for GARCH when  $\kappa_\eta > 3$ ?
- ▣ What does the variance ratio of GARCH-MIDAS measure?
- ▣ Why does stable convergence matter for the CLT of RV?
- ▣ Why can a signature plot rise at high frequency?
- ▣ Why is  $\beta_{dQ}$  expected to be negative in HARQ?
- ▣ Which losses keep the ranking under a noisy proxy?

### Next:

- ▣ VaR, ES and backtesting: elicibility, scoring and model risk
- ▣ From volatility forecasts to tail forecasts: VaR 1% and ES 2.5%, their scoring functions and backtests

## Appendix: the GARCH sandwich

- $\ell_t = -\frac{1}{2}(\ln \sigma_t^2 + \varepsilon_t^2/\sigma_t^2)$ ,  $d_t = \sigma_t^{-2} \partial \sigma_t^2 / \partial \theta$ :  $s_t = \frac{1}{2}(\eta_t^2 - 1)d_t$  at  $\theta_0$
- $B = \mathbb{E} s_t s_t' = \frac{1}{4} \mathbb{E}(\eta_t^2 - 1)^2 \mathbb{E} d_t d_t' = \frac{\kappa_\eta - 1}{4} J$ , because  $\eta_t$  is independent of  $d_t \in \mathcal{F}_{t-1}$
- $\partial s_t / \partial \theta' = -\frac{1}{2} \eta_t^2 d_t d_t' + \frac{1}{2}(\eta_t^2 - 1) \partial d_t / \partial \theta'$ ; the second term has mean zero, so  $A = \frac{1}{2} J$
- $A^{-1} B A^{-1} = 4J^{-1} \cdot \frac{\kappa_\eta - 1}{4} J \cdot J^{-1} = (\kappa_\eta - 1)J^{-1}$ ; Hessian only:  $A^{-1} = 2J^{-1}$ ; OPG:  $B^{-1} = \frac{4}{\kappa_\eta - 1} J^{-1}$

## Appendix: the noise bias of realised variance

- ▣  $\tilde{r}_i = Y_{i/n} - Y_{(i-1)/n} = r_i + u_i - u_{i-1}$ , with  $u$  i.i.d.  $(0, \omega^2)$  independent of  $X$
- ▣  $\sum_i \tilde{r}_i^2 = \sum_i r_i^2 + 2 \sum_i r_i(u_i - u_{i-1}) + \sum_i (u_i - u_{i-1})^2$ ; the cross term has mean zero, the last has mean  $2n\omega^2$
- ▣  $\text{Cov}(\tilde{r}_i, \tilde{r}_{i-1}) = -\omega^2$ : first-order negative autocorrelation; the realised kernel adds  $2\gamma_1 \approx -2n\omega^2$  back and removes the bias
- ▣ Positively autocorrelated returns (gradual price adjustment, stale prices) give  $\gamma_1 > 0$  and a signature plot that rises with the interval; the kernel corrects both signs

## Appendix: attenuation and HARQ

- ▣  $IV_{t+1} = c + \phi IV_t + v_{t+1}$ , observed  $RV_t = IV_t + e_t$ ,  $e_t$  uncorrelated with  $IV_t$  and  $v_{t+1}$
- ▣ OLS of  $RV_{t+1}$  on  $RV_t$ :  $\text{plim } \hat{\phi} = \frac{\text{Cov}(IV_{t+1}, IV_t)}{\text{Var}(IV_t) + \text{Var}(e_t)} = \phi\lambda$ ,  $\lambda = \frac{\text{Var}(IV)}{\text{Var}(IV) + \mathbb{E}\text{Var}(e_t)}$
- ▣ The best linear predictor given today's precision uses  $\phi\lambda_t$ ,  $\lambda_t = \frac{\text{Var}(IV)}{\text{Var}(IV) + 2IQ_t/n}$ : decreasing in  $IQ_t$
- ▣ A first-order expansion in  $\sqrt{IQ_t}$  gives  $\beta_d + \beta_{dQ}\sqrt{RQ_t}$  with  $\beta_{dQ} < 0$ : the HARQ specification [Bollerslev, Patton and Quaadvlieg \(2016\)](#)

## References (1/5)

- ▣ Aielli, G. P. (2013). Dynamic conditional correlation: On properties and estimation. *Journal of Business & Economic Statistics*, 31(3), 282–299. doi:10.1080/07350015.2013.771027
- ▣ Andersen, T. G., & Bollerslev, T. (1998). Answering the skeptics: Yes, standard volatility models do provide accurate forecasts. *International Economic Review*, 39(4), 885–905. doi:10.2307/2527343
- ▣ Andersen, T. G., Bollerslev, T., & Diebold, F. X. (2007). Roughing it up: Including jump components in the measurement, modeling, and forecasting of return volatility. *The Review of Economics and Statistics*, 89(4), 701–720. doi:10.1162/rest.89.4.701
- ▣ Andersen, T. G., Bollerslev, T., Diebold, F. X., & Labys, P. (2003). Modeling and forecasting realized volatility. *Econometrica*, 71(2), 579–625. doi:10.1111/1468-0262.00418
- ▣ Andersen, T. G., Dobrev, D., & Schaumburg, E. (2012). Jump-robust volatility estimation using nearest neighbor truncation. *Journal of Econometrics*, 169(1), 75–93. doi:10.1016/j.jeconom.2012.01.011
- ▣ Bandi, F. M., & Russell, J. R. (2008). Microstructure noise, realized variance, and optimal sampling. *The Review of Economic Studies*, 75(2), 339–369. doi:10.1111/j.1467-937X.2008.00474.x
- ▣ Barndorff-Nielsen, O. E., Hansen, P. R., Lunde, A., & Shephard, N. (2008). Designing realized kernels to measure the ex post variation of equity prices in the presence of noise. *Econometrica*, 76(6), 1481–1536. doi:10.3982/ECTA6495
- ▣ Barndorff-Nielsen, O. E., Hansen, P. R., Lunde, A., & Shephard, N. (2009). Realized kernels in practice: Trades and quotes. *The Econometrics Journal*, 12(3), C1–C32. doi:10.1111/j.1368-423X.2008.00275.x
- ▣ Barndorff-Nielsen, O. E., Hansen, P. R., Lunde, A., & Shephard, N. (2011). Multivariate realised kernels: Consistent positive semi-definite estimators of the covariation of equity prices with noise and non-synchronous trading. *Journal of Econometrics*, 162(2), 149–169. doi:10.1016/j.jeconom.2010.07.009
- ▣ Barndorff-Nielsen, O. E., & Shephard, N. (2002). Econometric analysis of realized volatility and its use in estimating stochastic volatility models. *Journal of the Royal Statistical Society: Series B*, 64(2), 253–280. doi:10.1111/1467-9868.00336
- ▣ Barndorff-Nielsen, O. E., & Shephard, N. (2004). Power and bipower variation with stochastic volatility and jumps. *Journal of Financial Econometrics*, 2(1), 1–37. doi:10.1093/jjfinec/nbh001
- ▣ Barndorff-Nielsen, O. E., & Shephard, N. (2006). Econometrics of testing for jumps in financial economics using bipower variation. *Journal of Financial Econometrics*, 4(1), 1–30. doi:10.1093/jjfinec/nbi022

## References (2/5)

- ▣ Berkes, I., Horváth, L., & Kokoszka, P. (2003). GARCH processes: Structure and estimation. *Bernoulli*, 9(2), 201–227. doi:10.3150/bj/1068128975
- ▣ Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3), 307–327. doi:10.1016/0304-4076(86)90063-1
- ▣ Bollerslev, T. (1990). Modelling the coherence in short-run nominal exchange rates: A multivariate generalized ARCH model. *The Review of Economics and Statistics*, 72(3), 498–505. doi:10.2307/2109358
- ▣ Bollerslev, T., Patton, A. J., & Quaedvlieg, R. (2016). Exploiting the errors: A simple approach for improved volatility forecasting. *Journal of Econometrics*, 192(1), 1–18. doi:10.1016/j.jeconom.2015.10.007
- ▣ Bollerslev, T., & Wooldridge, J. M. (1992). Quasi-maximum likelihood estimation and inference in dynamic models with time-varying covariances. *Econometric Reviews*, 11(2), 143–172. doi:10.1080/07474939208800229
- ▣ Chiriac, R., & Voev, V. (2011). Modelling and forecasting multivariate realized volatility. *Journal of Applied Econometrics*, 26(6), 922–947. doi:10.1002/jae.1152
- ▣ Comte, F., & Lieberman, O. (2003). Asymptotic theory for multivariate GARCH processes. *Journal of Multivariate Analysis*, 84(1), 61–84. doi:10.1016/S0047-259X(02)00009-X
- ▣ Conrad, C., & Kleen, O. (2020). Two are better than one: Volatility forecasting using multiplicative component GARCH-MIDAS models. *Journal of Applied Econometrics*, 35(1), 19–45. doi:10.1002/jae.2742
- ▣ Corsi, F. (2009). A simple approximate long-memory model of realized volatility. *Journal of Financial Econometrics*, 7(2), 174–196. doi:10.1093/jjfinec/nbp001
- ▣ Diebold, F. X., & Mariano, R. S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3), 253–263. doi:10.1080/07350015.1995.10524599
- ▣ Engle, R. (2002). Dynamic conditional correlation: A simple class of multivariate generalized autoregressive conditional heteroskedasticity models. *Journal of Business & Economic Statistics*, 20(3), 339–350. doi:10.1198/073500102288618487
- ▣ Engle, R. F. (1982). Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation. *Econometrica*, 50(4), 987–1007. doi:10.2307/1912773

## References (3/5)

- Engle, R. F., Ghysels, E., & Sohn, B. (2013). Stock market volatility and macroeconomic fundamentals. *The Review of Economics and Statistics*, 95(3), 776–797. doi:10.1162/REST\_a\_00300
- Engle, R. F., & Kroner, K. F. (1995). Multivariate simultaneous generalized ARCH. *Econometric Theory*, 11(1), 122–150. doi:10.1017/S0266466600009063
- Engle, R. F., Ledoit, O., & Wolf, M. (2019). Large dynamic covariance matrices. *Journal of Business & Economic Statistics*, 37(2), 363–375. doi:10.1080/07350015.2017.1345683
- Engle, R. F., & Lee, G. G. J. (1999). A long-run and short-run component model of stock return volatility. In R. F. Engle & H. White (Eds.), *Cointegration, Causality, and Forecasting: A Festschrift in Honour of Clive W. J. Granger* (pp. 475–497). Oxford University Press. doi:10.1093/oso/9780198296836.003.0020
- Engle, R. F., & Sheppard, K. (2001). Theoretical and empirical properties of dynamic conditional correlation multivariate GARCH. *NBER Working Paper* 8554. doi:10.3386/w8554
- Epps, T. W. (1979). Comovements in stock prices in the very short run. *Journal of the American Statistical Association*, 74(366), 291–298. doi:10.1080/01621459.1979.10482508
- Francq, C., & Zakoian, J.-M. (2004). Maximum likelihood estimation of pure GARCH and ARMA-GARCH processes. *Bernoulli*, 10(4), 605–637. doi:10.3150/bj/1093265632
- Francq, C., & Zakoian, J.-M. (2019). *GARCH Models: Structure, Statistical Inference and Financial Applications* (2nd ed.). Wiley. doi:10.1002/9781119313472
- Ghysels, E., Santa-Clara, P., & Valkanov, R. (2006). Predicting volatility: Getting the most out of return data sampled at different frequencies. *Journal of Econometrics*, 131(1–2), 59–95. doi:10.1016/j.jeconom.2005.01.004
- Glosten, L. R., Jagannathan, R., & Runkle, D. E. (1993). On the relation between the expected value and the volatility of the nominal excess return on stocks. *The Journal of Finance*, 48(5), 1779–1801. doi:10.1111/j.1540-6261.1993.tb05128.x
- Hafner, C. M., & Preminger, A. (2009). On asymptotic theory for multivariate GARCH models. *Journal of Multivariate Analysis*, 100(9), 2044–2054. doi:10.1016/j.jmva.2009.03.011
- Hamilton, J. D. (1994). *Time Series Analysis*. Princeton University Press. doi:10.2307/j.ctv14jx6sm

## References (4/5)

- Hansen, P. R., & Huang, Z. (2016). Exponential GARCH modeling with realized measures of volatility. *Journal of Business & Economic Statistics*, 34(2), 269–287. doi:10.1080/07350015.2015.1038543
- Hansen, P. R., Huang, Z., & Shek, H. H. (2012). Realized GARCH: A joint model for returns and realized measures of volatility. *Journal of Applied Econometrics*, 27(6), 877–906. doi:10.1002/jae.1234
- Hansen, P. R., & Lunde, A. (2005). A forecast comparison of volatility models: Does anything beat a GARCH(1,1)? *Journal of Applied Econometrics*, 20(7), 873–889. doi:10.1002/jae.800
- Hansen, P. R., & Lunde, A. (2006a). Consistent ranking of volatility models. *Journal of Econometrics*, 131(1–2), 97–121. doi:10.1016/j.jeconom.2005.01.005
- Hansen, P. R., & Lunde, A. (2006b). Realized variance and market microstructure noise. *Journal of Business & Economic Statistics*, 24(2), 127–161. doi:10.1198/073500106000000071
- Hansen, P. R., Lunde, A., & Nason, J. M. (2011). The model confidence set. *Econometrica*, 79(2), 453–497. doi:10.3982/ECTA5771
- Heber, G., Lunde, A., Shephard, N., & Sheppard, K. (2009). *Oxford-Man Institute's realized library*, version 0.3. Oxford-Man Institute, University of Oxford (discontinued in 2022; copy of 28 February 2022 preserved by the Internet Archive). [web.archive.org](http://web.archive.org)
- Huang, X., & Tauchen, G. (2005). The relative contribution of jumps to total price variance. *Journal of Financial Econometrics*, 3(4), 456–499. doi:10.1093/jjfinec/nbi025
- Jacod, J., Li, Y., Mykland, P. A., Podolskij, M., & Vetter, M. (2009). Microstructure noise in the continuous case: The pre-averaging approach. *Stochastic Processes and their Applications*, 119(7), 2249–2276. doi:10.1016/j.spa.2008.11.004
- Jensen, S. T., & Rahbek, A. (2004). Asymptotic inference for nonstationary GARCH. *Econometric Theory*, 20(6), 1203–1226. doi:10.1017/S0266466604206065
- Laurent, S., Rombouts, J. V. K., & Violante, F. (2012). On the forecasting accuracy of multivariate GARCH models. *Journal of Applied Econometrics*, 27(6), 934–955. doi:10.1002/jae.1248
- Ledoit, O., & Wolf, M. (2004). A well-conditioned estimator for large-dimensional covariance matrices. *Journal of Multivariate Analysis*, 88(2), 365–411. doi:10.1016/S0047-259X(03)00096-4

## References (5/5)

- Ledoit, O., & Wolf, M. (2020). Analytical nonlinear shrinkage of large-dimensional covariance matrices. *The Annals of Statistics*, 48(5), 3043–3065. doi:10.1214/19-AOS1921
- Lee, S. S., & Mykland, P. A. (2008). Jumps in financial markets: A new nonparametric test and jump dynamics. *The Review of Financial Studies*, 21(6), 2535–2563. doi:10.1093/rfs/hhm056
- Lee, S.-W., & Hansen, B. E. (1994). Asymptotic theory for the GARCH(1,1) quasi-maximum likelihood estimator. *Econometric Theory*, 10(1), 29–52. doi:10.1017/S0266466600008215
- Ling, S., & McAleer, M. (2003). Asymptotic theory for a vector ARMA-GARCH model. *Econometric Theory*, 19(2), 280–310. doi:10.1017/S0266466603192092
- Lumsdaine, R. L. (1996). Consistency and asymptotic normality of the quasi-maximum likelihood estimator in IGARCH(1,1) and covariance stationary GARCH(1,1) models. *Econometrica*, 64(3), 575–596. doi:10.2307/2171862
- Newey, W. K., & West, K. D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3), 703–708. doi:10.2307/1913610
- Pakel, C., Shephard, N., Sheppard, K., & Engle, R. F. (2021). Fitting vast dimensional time-varying covariance models. *Journal of Business & Economic Statistics*, 39(3), 652–668. doi:10.1080/07350015.2020.1713795
- Patton, A. J. (2011). Volatility forecast comparison using imperfect volatility proxies. *Journal of Econometrics*, 160(1), 246–256. doi:10.1016/j.jeconom.2010.03.034
- Patton, A. J., & Sheppard, K. (2015). Good volatility, bad volatility: Signed jumps and the persistence of volatility. *The Review of Economics and Statistics*, 97(3), 683–697. doi:10.1162/REST\_a\_00503
- Petropoulos, F., Apiletti, D., Assimakopoulos, V., Babai, M. Z., et al. (2022). Forecasting: Theory and practice. *International Journal of Forecasting*, 38(3), 705–871. doi:10.1016/j.ijforecast.2021.11.001
- Shephard, N., & Sheppard, K. (2010). Realising the future: Forecasting with high-frequency-based volatility (HEAVY) models. *Journal of Applied Econometrics*, 25(2), 197–231. doi:10.1002/jae.1158
- Zhang, L., Mykland, P. A., & Aït-Sahalia, Y. (2005). A tale of two time scales: Determining integrated volatility with noisy high-frequency data. *Journal of the American Statistical Association*, 100(472), 1394–1411. doi:10.1198/016214505000000169